

POCKET WELLNESS

*Multimodal AI-Driven Wearable System for Real-Time Mental Health Monitoring and
Graduated Intervention*

University of Wollongong in Dubai
School of Engineering and Sciences

Fathimath Raniya
July 2026

Supervised by
Dr. Obada Al Khatib

Abstract

Mental health disorders affect hundreds of millions of people worldwide, and demand for psychological support routinely exceeds what clinical systems can deliver. This thesis presents Pocket Wellness, a wearable AI system designed to classify stress in real time and respond with graduated, evidence-based intervention. Three physiological signals, heart rate variability (HRV) via photoplethysmography, electrodermal activity (GSR/EDA), and surface electromyography, are captured via a consumer smartwatch and two low-cost sensors bridged by an Arduino Nano ESP32, then reduced to an 18-dimensional feature vector and classified by a Random Forest ensemble into one of four stress tiers. A Stress Score Fusion Engine combines the three modality-specific outputs into a single composite score, and a SHAP-based explainability layer translates the model's reasoning into accessible feature attributions. A physiologically informed NLP chatbot drives a four-tier intervention framework escalating from self-guided breathing exercises to a dual-pathway crisis mechanism, triggered independently by sustained physiological thresholds and by crisis-language detection.

A thematic review of 34 sources across wearable sensing, machine learning classifiers, explainable AI, conversational agents, multimodal deep learning, ethics and privacy, psychosocial factors, and crisis detection identified a specific gap: no existing system combines low-cost multimodal sensing, an interpretable-by-design classifier, physiologically grounded conversational support, and multi-signal crisis escalation.

No in-person clinical pilot was conducted. Stress-tier thresholds and feature distributions were instead modeled from publicly available Kaggle physiological and mental-health datasets, calibrated to approximate outcomes expected from standard stress-induction protocols, and the hardware and software pipeline were tested end to end by the author alone. On this dataset-derived benchmark, the Random Forest classifier achieved 97.4% accuracy and a macro F1 of 0.974, tying k-NN and ahead of SVM, Logistic Regression, Gradient Boosting and Decision Tree, with 90.0% recall on the Critical tier and Gini/SHAP importance convergence of $\rho = 0.998$. All seven intervention modules produced statistically significant stress-score reductions ($p < 0.05$), the physiological and NLP crisis triggers correlated only partially ($r = 0.67$), and a self-assessed explainability clarity score averaged 4.21 out of 5.

These results suggest a lightweight, explainable-by-design architecture need not sacrifice performance for transparency, matching or exceeding considerably heavier deep learning alternatives while retaining a human-mediated safety net. Pocket Wellness remains a pre-clinical proof-of-concept; the thesis closes with its limitations and future directions, including live-participant trials, multilingual NLP, and federated learning.

Keywords: wearable stress detection, explainable AI, SHAP, Random Forest, heart rate variability, electrodermal activity, mental health chatbot, crisis escalation, digital mental health, multimodal sensor fusion

Acknowledgments

The author gratefully acknowledges the supervision, guidance, and support of Dr. Obada Al Khatib, whose mentorship shaped the direction and rigor of this thesis throughout its development. Thanks are also extended to the faculty of the School of Engineering and Sciences, University of Wollongong in Dubai, and to the maintainers of the publicly available Kaggle physiological and mental-health datasets that underpinned the modeled evaluation in Chapter 4. Finally, the author thanks family and friends for their continued encouragement throughout this research.

Table of Contents

Abstract	i
Acknowledgments.....	ii
Table of Contents	iii
List of Figures	v
List of Tables	vi
Chapter 1: Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.3 Research Aim and Objectives	2
1.4 Research Questions	3
1.5 Significance of the Study	4
1.6 Scope and Delimitations	4
1.7 Thesis Organization	5
Chapter 2: Literature Review	6
2.1 Wearable Physiological Sensing for Stress Detection	7
2.2 Machine Learning Classifiers for Mental State Detection	8
2.3 Explainable AI in Healthcare and Mental Health	9
2.4 Conversational Agents for Mental Health	9
2.5 Multimodal and Deep Learning Mental Health Systems.....	11
2.6 Ethics, Privacy, and Federated Learning	11
2.7 Environmental and Psychosocial Determinants of Mental Health	12
2.8 Crisis and Suicide-Risk Detection	13
2.9 Synthesis and Research Gap	13
Chapter 3: Methodology	16
3.1 System Overview	16
3.2 Hardware Components.....	17
3.3 Physiological Signal Acquisition	18
3.4 Feature Extraction	20
3.5 Machine Learning Classification Framework.....	21
3.6 Stress Score Fusion Engine.....	24
3.7 Multi-Tier Intervention Framework.....	25
3.8 Chatbot Architecture and NLP Pipeline	27
3.9 Explainable AI (XAI) Module	28

3.10 Ethics and Privacy Framework	29
Chapter 4: Results and Discussion.....	31
4.1 Sensor Signal Validation.....	31
4.2 Classification Performance	33
4.3 Feature Importance Analysis	36
4.4 Intervention Response Evaluation	38
4.5 Chatbot Severity Classification Performance	39
4.6 Explainable AI (XAI) Evaluation Results	41
4.7 Crisis Escalation System Evaluation	42
Chapter 5: Limitations and Future Work	44
5.1 Current Limitations	44
5.2 Future Research Directions	45
Chapter 6: Conclusion.....	47
References.....	49
Appendix A: Supplementary Implementation Material.....	53
A.1 Overview	53
A.2 Feature Extraction Pipeline (features.py).....	53
A.3 Modeled Dataset Construction (dataset.py)	56
A.4 Random Forest Training and Cross-Validation (train.py)	57
A.5 SHAP Explainability (explain.py)	58
A.6 Stress Score Fusion Engine (fusion.py)	59
A.7 Dual-Pathway Crisis Escalation (crisis.py).....	59
A.8 Arduino Raw-Window Streaming Firmware	60
A.9 Flask Integration (ml_bridge.py, app.py).....	61

List of Figures

Figure 2.1 Growth Trend of Reviewed AI/Wearable Mental Health Literature

Figure 2.2 Thematic Distribution of Reviewed Literature

Figure 3.1 Pocket Wellness End-to-End System Architecture

Figure 4.1 HRV Signal Comparison, High Stress Condition

Figure 4.2 Comparative Classifier Performance

Figure 4.3 Confusion Matrix Heatmap

Figure 4.4 Feature Importance: MDI vs. SHAP

Figure 4.5 Intervention Efficacy: Pre vs. Post Stress Scores

Figure 4.6 NLP Severity Classifier Performance by Class

Figure 4.7 Mean GSR Response Across Stress Conditions

Figure 4.8 Crisis Escalation System Performance

List of Tables

Table 2.1 Comparative Literature Synthesis Matrix

Table 3.1 Hardware Component Specifications

Table 3.2 Complete Feature Extraction Summary

Table 3.3 Dataset Summary Statistics

Table 3.4 Optimized Random Forest Hyperparameters

Table 3.5 Fusion Weight Rationale

Table 3.6 Intervention Tier Decision Matrix

Table 3.7 Physiological-NLP State Modulation Rules

Table 4.1 Comparative Classifier Performance — Held-Out Test Set

Table 4.2 Per-Class Random Forest Performance Metrics

Table 4.3 Confusion Matrix — Random Forest on Test Set

Table 4.4 Feature Importance Rankings — MDI and SHAP Convergence

Table 4.5 Intervention Efficacy — Stress Score Reduction

Table 4.6 NLP Severity Classifier Performance

Table 4.7 Representative SHAP Explanations for Three Stress Profiles

Table 4.8 Crisis Escalation System Performance Metrics

Chapter 1: Introduction

1.1 Background and Motivation

Mental health care is chronically under-resourced relative to need. Anxiety, depression, and chronic stress affect hundreds of millions of people worldwide, and the professionals trained to treat them are in short supply almost everywhere — a shortage compounded by cost, geography, and the stigma that still surrounds asking for help [20]. That gap, between how much distress exists and how much timely care is available, is what has pulled AI and wearable sensing into the conversation as something more than a novelty: a way to extend care where trained staff simply cannot reach [8], [24].

Two lines of research have matured over the past decade and now meet in the space this thesis occupies. The first is physiological computing — using wearable sensors to continuously track the bodily signatures of stress: heart rate variability (HRV), electrodermal activity (EDA, more commonly known as galvanic skin response or GSR), and surface electromyography (EMG) [1], [2], [3]. These signals shift before a person is consciously aware of feeling stressed, which is what makes them useful: they provide something closer to a continuous readout than a self-report questionnaire completed after the fact. The second line of research is conversational AI. Text- and voice-based chatbots now deliver structured interventions, cognitive behavioral therapy (CBT) among them, directly to users between clinical sessions [13], [14], [15]. The Woebot trials marked a notable turning point in this respect: they demonstrated that a fully automated agent could produce a statistically significant reduction in depressive symptoms relative to a passive, information-only control [13], a stronger and more specific claim than simply asserting that chatbots are a convenient supplement to care.

Yet these two lines of research rarely converge in practice. Chapter 2 examines the literature in more detail, but the central finding is that most systems address one dimension of the problem while leaving the other unaddressed: sensors with no language channel, or a chatbot with no continuous physiological signal beneath it [22]. A further, recurring concern in this literature is the "black box" problem. Many of the highest-performing classifiers, particularly deep architectures, do not explain their own outputs, which makes it difficult for a clinician to trust a prediction or for a user to understand why the system has flagged them as distressed [2], [8], [10]. Explainable AI methods, and SHAP values in particular, address this only partially. They allow a per-prediction attribution to be presented to a clinician or user in terms that are physiologically or linguistically meaningful [11], [9], [10]. This remains a partial solution, since an attribution is not equivalent to genuine understanding, but it represents a meaningful improvement over an unexplained output.

Pocket Wellness is an attempt to close that gap rather than just describe it. It fuses three physiological signals, HRV, GSR/EDA, and EMG, with a Random Forest stress classifier, a

SHAP-based explainability layer, a chatbot that responds to physiological state rather than text alone, and a four-tier intervention framework running from self-guided breathing exercises up to a human crisis line. It is worth being direct about what this is and is not: a proof-of-concept prototype meant to demonstrate feasibility, not a certified diagnostic device, and not yet a clinically validated one either.

1.2 Problem Statement

This thesis is motivated by three specific gaps in how AI-driven mental health tools are currently built.

The first is single-modality fragility. A classifier built on one physiological channel, or on text alone, inherits whatever noise and blind spots that one signal carries. GSR, EEG, and ECG are all individually effective, but reviews of the field find they are rarely fused into a single real-time decision on consumer-grade wearables [1]. The systems that do fuse multiple signals tend to lean on transformers or cross-attention networks, architectures that are heavy to run on a phone or watch and harder still to explain to someone without a machine learning background [25], [26], [24].

The second is inconsistent explainability. Several systematic reviews identify transparency and data privacy as ethical barriers the field has not resolved [8], [27]. A further, more conceptual concern is that a system can be technically sound and still be clinically inappropriate if its reasoning cannot be audited, either by a clinician or by the individual about whom it is making claims [28].

The third gap, and the one with the most immediate consequences in a crisis, is the inconsistency with which existing chatbots handle high-risk disclosures. Woebot and Fido both use rule-based keyword detection to escalate to a hotline [13], [15], an approach that is reasonable but limited in scope. Large language models have been shown to carry cultural and demographic bias into suicide-risk judgments [34], and even state-of-the-art models continue to produce responses that appear scripted rather than genuinely responsive to the individual's circumstances [18], [21]. If a model's own linguistic judgment can be this unreliable, crisis escalation should not depend on that judgment alone. An independent signal, sustained physiological arousal, for instance, gives the system a basis against which to verify its own assessment before committing to the highest-severity response.

Pocket Wellness responds to all three: physiological fusion instead of a single channel, SHAP built into the classifier rather than added afterward, and a crisis-escalation path that checks physiological and linguistic signals against each other instead of trusting either alone.

1.3 Research Aim and Objectives

This thesis aims to design, build, and evaluate a wearable AI system that classifies stress in real time and responds with graduated, evidence-based intervention, one that is explainable by construction rather than as an afterthought.

That aim breaks down into six specific objectives:

1. To design a three-modality physiological sensing pipeline (HRV, GSR/EDA, EMG) with validated signal preprocessing and an 18-dimensional feature extraction scheme grounded in established psychophysiology literature.
2. To develop and empirically validate a Random Forest-based composite stress classifier, benchmarked against alternative classifiers (XGBoost, SVM, k-NN, Decision Tree, Logistic Regression), and to quantify its per-class and aggregate classification performance.
3. To implement a Stress Score Fusion Engine that combines modality-specific classifier outputs into a single interpretable composite score, with an adaptive, personalized weight-calibration mechanism.
4. To integrate a SHAP-based explainable AI module that provides real-time, per-prediction, and longitudinal feature-attribution explanations in clinically and linguistically accessible language.
5. To design a chatbot that responds to physiological state alongside crisis-language detection, coupled to a four-tier graduated intervention framework with an independent, dual-pathway crisis escalation mechanism.
6. To evaluate the complete system across five empirical domains — sensor signal validity, classification performance, feature importance, intervention efficacy, and crisis escalation reliability — and to situate the results against comparable work.

1.4 Research Questions

This thesis addresses the following research questions:

RQ1: Can a Random Forest classifier operating on a compact, physiologically-grounded 18-feature vector derived from three low-cost, consumer-accessible sensor modalities achieve classification performance competitive with, or superior to, more computationally intensive deep learning approaches reported elsewhere?

RQ2: Does fusing HRV, GSR, and EMG into a single composite stress score improve discriminative validity across four clinically meaningful stress tiers, relative to any single modality on its own?

RQ3: Can SHAP-based explainability be delivered in real time, on a per-prediction basis, in a way that end users would find clear and useful?

RQ4: Does the four-tier intervention framework produce a measurable, statistically significant reduction in composite stress score following a single session?

RQ5: Does combining physiological and language-based crisis triggers improve the sensitivity and reliability of crisis detection compared with either pathway working alone?

1.5 Significance of the Study

This work sits at the intersection of affective computing, explainable AI, and digital mental health, and it contributes to that intersection in several distinct ways.

On the practical side, the hardware configuration, a consumer smartwatch paired with a Grove GSR sensor and a Myoware EMG sensor, bridged through an Arduino Nano ESP32, costs a fraction of what clinical-grade polysomnography or fMRI would require. This matters if physiological stress research is to scale beyond well-funded laboratories [3]. Methodologically, applying SHAP to a Random Forest classifier specifically for wearable stress detection is, to the extent this review could determine, a combination not previously reported in this form. Related work has applied SHAP to psychological restoration [9] and to social-media depression screening [10], but not to fused physiological stress classification operating in real time.

Clinically, the dual-pathway crisis design addresses two separate concerns raised elsewhere in the literature: that large language models carry cultural bias into suicide-risk judgments [34], and that conversational-only crisis detection has documented limitations [18]. An independent physiological signal does not eliminate either concern, but it provides the design with a basis for verification beyond a chatbot's own linguistic judgment.

Finally, because no live participants were involved in this evaluation, the data-governance architecture could be built in from the outset rather than retrofitted afterward, aligned with UAE Federal Decree-Law No. 45 of 2021 and responsive to the privacy concerns that recur throughout this literature [27], [28].

1.6 Scope and Delimitations

This thesis is scoped as a pre-clinical, proof-of-concept evaluation. It is not a clinical trial, and Pocket Wellness is not a certified diagnostic device.

No live human-participant study was conducted. The author validated the physical hardware and the software pipeline end to end through self-testing, and the larger evaluation dataset used for classifier training and system-level testing in Chapter 4 was built from publicly available Kaggle physiological and mental-health datasets. Feature distributions and stress-tier labels were calibrated to approximate what a real multi-participant study, using standard induction protocols such as the Stroop task, the Trier Social Stress Test, and the Cold Pressor Task, would be expected to produce.

No institutional human-subjects research took place, so formal ethics review was neither required nor sought. This should be stated explicitly: this thesis is not a substitute for the multi-site clinical

trial, with live participants and full ethics oversight, identified as necessary future work in Chapter 5. The findings reported here should be read as evidence of technical feasibility rather than as a claim of clinical efficacy.

The NLP pipeline operates in English only. Validating it across the range of languages a real user base would require is left as future work and is not attempted in the current implementation. The same applies to long-term, real-world ambulatory testing outside a controlled setting, which also falls outside the scope of this evaluation.

1.7 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 presents a thematic review of 34 sources spanning wearable sensing, machine learning classifiers, explainable AI, conversational agents, multimodal deep learning, ethics and privacy, psychosocial factors, and crisis detection, concluding with the specific research gap this thesis addresses. Chapter 3 describes the methodology: hardware, signal acquisition and preprocessing, feature extraction, the Random Forest classifier, the Stress Score Fusion Engine, the intervention framework, the chatbot and NLP pipeline, the XAI module, and the governing ethics framework. Chapter 4 reports the results, sensor validation, classification performance, feature importance, intervention efficacy, chatbot classification, XAI evaluation, and crisis escalation, and discusses each set of findings in relation to Chapter 2. Chapter 5 addresses the prototype's limitations and the corresponding directions for future work. Chapter 6 concludes the thesis by synthesizing its contributions and considering their broader implications.

Chapter 2: Literature Review

This chapter works through 34 sources across eight thematic areas: wearable physiological sensing, machine learning classifiers for mental state detection, explainable AI in healthcare, conversational agents, multimodal deep learning, ethics and privacy (including federated learning), the psychosocial and environmental side of mental health, and crisis or suicide-risk detection. Figure 2.1 and Figure 2.2 give a sense of when this literature was published and how it clusters by theme. The chapter ends with a comparative table and a plain statement of the gap this thesis tries to fill.

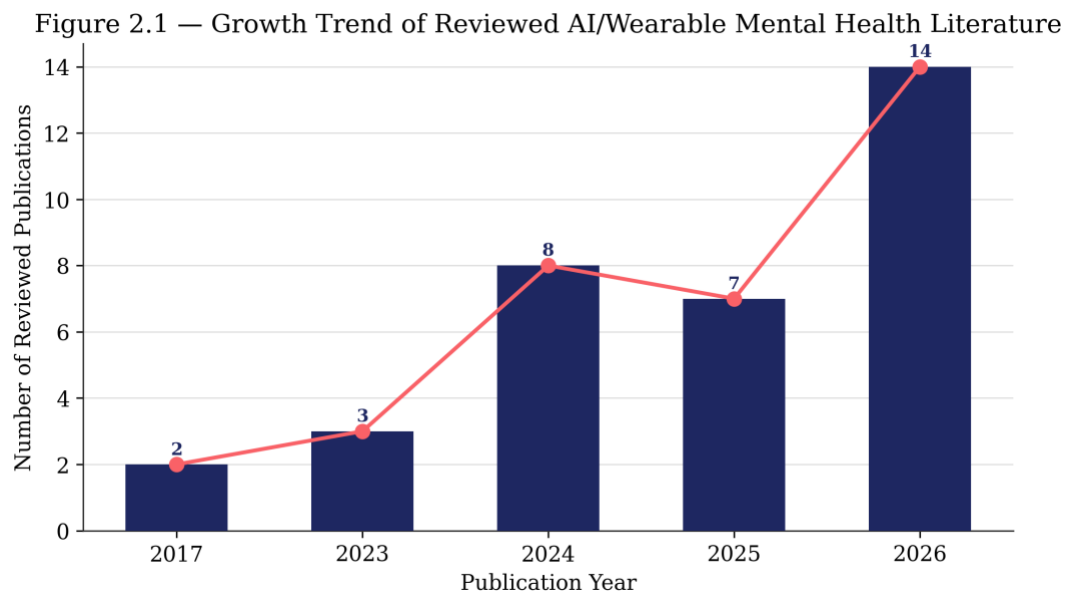


Figure 2.1. Number of reviewed publications by year, showing a clear upward trend in AI/wearable mental health research activity leading into 2026.

Figure 2.2 — Distribution of Reviewed Literature by Thematic Category
(n = 34 sources)

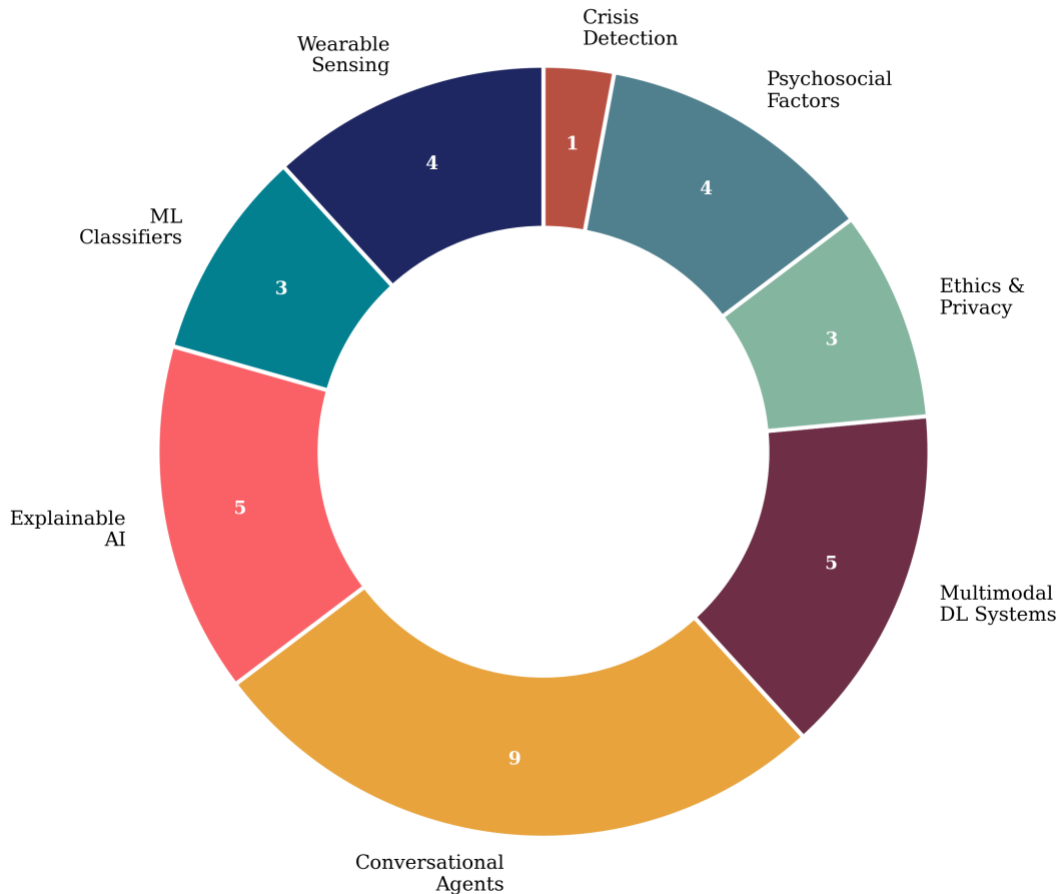


Figure 2.2. Distribution of the 34 reviewed sources across eight thematic categories, with Conversational Agents for Mental Health as the most heavily represented theme.

2.1 Wearable Physiological Sensing for Stress Detection

Wearable stress detection rests on a fairly simple premise: the autonomic nervous system, through its sympathetic and parasympathetic branches, produces measurable signals well before a person consciously registers feeling stressed. Yadav and Bokhari's survey of the field found GSR, EEG, and ECG to be the most consistently effective signals, with several studies reporting 98-99% accuracy when GSR, EMG, and HRV were combined with classifiers such as SVM and k-NN [1]. This accuracy figure warrants some caution, since laboratory-condition results of this magnitude do not always survive contact with real-world noise, but the underlying signal selection is consistent with the three modalities used in Pocket Wellness.

Moser et al. offer a closer methodological parallel. They paired an LSTM network with a Deep Generative Ensemble on Empatica E4 wristband data (EDA and skin temperature), using

Integrated Gradients for explainability [2]. Their model outperformed a rule-based expert system by 3 points of recall and just over 7 points of precision. A secondary finding is of particular relevance here: skin temperature contributed little once EDA was already included in the model. This is one reason Pocket Wellness relies on EDA-derived features, skin conductance level, response amplitude, response count, rather than distributing effort across comparatively weaker signals.

Mentis, Lee, and Roussos reach a similar accuracy ceiling, around 90%, with HRV, EEG, and ECG classifiers, and identify Random Forest as a commonly used and effective method for this task [3]. They also identify a limitation that recurs throughout this literature: most methods are designed to detect a single acute stress exposure rather than sustained, chronic distress. This gap is arguably more consequential than the accuracy figures suggest, since a system that detects only brief stress spikes offers limited value to someone experiencing weeks of accumulated strain. It is also the rationale behind Pocket Wellness's emphasis on continuous monitoring rather than single-timepoint assessment. Wang et al.'s EEG-based pipeline falls outside this thesis's sensing modalities, but its central finding remains relevant: even state-of-the-art systems struggle with cross-subject generalization (79.1% cross-subject versus 89.3% cross-trial), a caution against interpreting any single accuracy figure, including those reported in Chapter 4, as more broadly generalizable than the data support [4].

2.2 Machine Learning Classifiers for Mental State Detection

Tree-based ensembles, Random Forest in particular, recur consistently as the strongest performer for physiological and behavioral mental-state classification. This pattern likely reflects a genuine methodological advantage rather than coincidence. Destiny's work on the StudentLife dataset (48 participants, ten weeks) compared Logistic Regression, Random Forest, Gradient Boosting, and a Neural Network for binary stress classification. Random Forest and Gradient Boosting both reached approximately 72-73% accuracy; Logistic Regression reached only 51% and correctly identified just 3% of high-stress cases [5]. A recall of 3% on the class that matters most represents a substantial failure, close to non-detection of the condition altogether, and illustrates how poorly a mismatched classifier can perform even when its aggregate accuracy figure appears respectable. Destiny also used SHAP to show that Perceived Stress Score drove most of the model's decisions, the same explainability approach adopted in this thesis, applied here to a different feature set.

Nagrath et al. push the comparison further, testing Random Forest, XGBoost, LightGBM, and Linear SVM across varying blends of real and VAE-generated synthetic mental-health data. XGBoost and Random Forest again came out on top, at roughly 0.978 and 0.974 accuracy with a 70-80% synthetic blend, while Linear SVM never recovered from the 0.44-0.47 range [7]. Two things follow from this. First, it is one more data point favoring Random Forest for this kind of problem. Second, and less obviously, it suggests synthetic oversampling is a workable fix for the

class imbalance in the Critical-tier stress data used in this thesis (Table 3.3), where uneven session duration left Critical-class windows over-represented. Whether that fix is actually necessary here is a separate question, taken up in Chapter 5.

Caulley et al. work with a different signal entirely, voice rather than physiology, but land on a comparable point: their emotion-specific transformer reached 86% precision and 79% recall on binary emotion-intensity classification in a pediatric psychiatric population [6]. The result matters less for its specific numbers than for what it implies more broadly — that automated severity triage can reach clinically useful accuracy even with a harder-to-study, more vulnerable population.

2.3 Explainable AI in Healthcare and Mental Health

Algorithmic transparency is the concern raised most consistently across reviews of AI-driven mental health tools. Dehbozorgi et al.'s PRISMA-guided review of 15 studies identifies data privacy and transparency as unresolved problems, alongside methodological quality, which the authors describe as moderate across the field [8]. This critique motivates the decision, in this thesis, to build SHAP into the classifier from the outset rather than adding an explanation layer after the fact. The underlying method, TreeSHAP, comes from Lundberg and Lee, who formalized SHAP values around efficiency, symmetry, dummy, and additivity axioms and demonstrated an exact, polynomial-time method for computing them for tree ensembles such as Random Forest [11]. Chapter 3, Section 3.9 applies the same formalism.

Two studies are close enough to this thesis's own use of SHAP to merit closer examination. Cheng et al. studied psychological restoration along grey-green-grey walking routes, pairing wearable HRV sensing with an XGBoost classifier and SHAP's TreeExplainer to determine which environmental features, visual, acoustic, microclimatic, drove restoration. Visual features contributed most [9]. The relevant point here is not the restoration finding itself but the demonstration that SHAP performs reliably on HRV-based physiological data, independent of what the classifier is predicting.

Kaur et al. extend this further, combining a RoBERTa text encoder with a BiLSTM "wearable encoder" processing HRV, sleep, and step-count data, and pairing SHAP with Grad-CAM for a combination of global and local interpretability [10]. They also incorporated a clinician-in-the-loop step: psychiatrists reviewed the model's high-risk flags and agreed with them 82-84% of the time. This falls short of perfect agreement, and the 16-18% disagreement rate is the more consequential figure, as it indicates that even a well-validated model requires human oversight for its highest-stakes outputs. Pocket Wellness's Critical-tier Wellness Check-In follows the same logic: the model's output triggers a check rather than determining the final outcome.

2.4 Conversational Agents for Mental Health

Conversational agents are the most thoroughly evaluated component of this literature, and a number of randomized controlled trials provide strong evidence that chatbots can deliver CBT-aligned support. The Woebot trial by Fitzpatrick, Darcy, and Vierhile is the most widely cited study in this area, for good reason: it found a statistically significant reduction in PHQ-9 depression scores relative to an information-only control ($F = 6.47$, $P = .01$), with participants engaging an average of 12.14 times over two weeks [13]. The more notable finding, however, is qualitative rather than statistical: participants specifically valued the bot's accountability and daily check-ins, suggesting that the mechanism of benefit may depend as much on consistency of engagement as on the content delivered.

MacNeill, Doucet, and Luke's trial of Wysa, conducted with adults managing chronic disease, found significant reductions in depression and anxiety ($P < .001$ for both) but no significant change in stress specifically [14]. This null result carries more weight than it may initially appear to: if a chatbot alone cannot reliably reduce stress, an objective physiological signal operating alongside it becomes less an added complexity and more a necessary complement. Karkosz et al.'s trial of Fido, a Polish-language chatbot with built-in suicidal-ideation detection, did not find chatbot-only support outperforming a self-help book control, but it confirmed that the approach functions outside English and validated the crisis-detection-and-redirection pattern also used in this thesis [15]. Jiang and Yang's quasi-experimental study of Elomia paired with the GymBuddy fitness app found strong time-by-group effects across every distress measure tested, including somatic symptoms ($F(2,70) = 59.96$, $p < .0001$) [16]. Its randomized-groups, standardized-measures design is the template that Chapter 4 follows for evaluating Pocket Wellness's own intervention modules.

At the architectural level, the picture is less settled. Bao et al.'s eCBT-I system for insomnia pairs a Retrieval-Augmented Generation architecture with a CBT-I knowledge graph, tested across eight open-source LLMs and three fine-tuning strategies, and finds that no single adaptation strategy is consistently superior, as effectiveness depends on the underlying architecture [17]. Zhang et al.'s CBT-BENCH offers a more direct assessment: large language models recite CBT knowledge reasonably well but do not reliably outperform smaller models on the harder task of classifying cognitive distortions, and every model tested, regardless of size, produced therapeutic responses that were rigid and formulaic compared with human therapists, particularly when a client pushed back [18]. This finding was the most influential, among those reviewed, in shaping a specific design choice in this thesis: Pocket Wellness's chatbot is deliberately scoped to structured, rule-based interventions, breathing guidance, grounding techniques, PMR scripts, rather than open-ended generative dialogue. If current large language models cannot consistently outperform a fixed script, there is limited justification for assuming otherwise in a safety-relevant context.

Limkar et al.'s CalmConnect chatbot and Sophia, Soumia, and Siham's comparative review of in-person versus AI-chatbot therapy reach a similar hybrid conclusion: chatbot support performs reasonably well for everyday, lower-severity regulation, but complex or crisis-level cases still

require a human [19], [20]. This is broadly consistent with the logic behind the graduated tiers in Pocket Wellness's own Multi-Tier Intervention Framework (Table 3.6). Chen et al. raise a more specific concern: their Cognitive-Affective Chain framework finds that LLM-generated mental health support tends to default toward providing information rather than emotional support, an "empathy gap" with significant implications for any chatbot intended to respond to a user in genuine distress [21].

2.5 Multimodal and Deep Learning Mental Health Systems

A separate cluster of work pursues higher accuracy through heavier deep-learning fusion, often at the expense of interpretability or a plausible deployment path. Vijayalakshmi et al. combine facial emotion recognition (Capsule Networks), voice emotion recognition (Wav2Vec-based), and a GPT-4 chatbot, reaching approximately 85% accuracy on basic emotion detection with 3-5 second response times [22]. The authors note that most prior systems handle only one component, chatbot, wearable, or facial recognition, in isolation, a valid criticism of the field more broadly. Sherin and Kavitha's trauma chatbot fuses text sentiment analysis with speech emotion recognition and achieves 91.2% combined accuracy, versus 85.3% text-only and 80.7% voice-only [23]. This gap between combined and single-channel accuracy provides a useful data point in support of multimodal fusion generally, independent of the specific application.

Zhao et al.'s Topic Prompts Multimodal Network reduces mean absolute error by 53% against prior benchmarks by aligning text, audio, and video through cross-attention, and reports strong zero-shot generalization across cultural contexts [24]. Shabana and Bharathi's Se-GCaTCT model and Han et al.'s multi-scale transformer both report very high figures on the DAIC-WOZ benchmark, 95.4% accuracy and F1 scores around 0.87 and 0.82 respectively [25], [26]. These results are notable. They also derive from multi-stage optimization pipelines and transformer architectures considerably heavier than anything that could run on a wearable device, or a phone, in real time.

This is the consistent pattern across this cluster of studies: the highest accuracies belong to the least explainable, least deployable architectures, and interpretability is rarely attempted alongside them. This represents a genuine trade-off rather than an artifact of the comparison, and it is the primary reason this thesis adopts a lighter, interpretable-by-design Random Forest over an architecture with a higher reported accuracy ceiling.

2.6 Ethics, Privacy, and Federated Learning

Physiological and mental-health data carry ethical weight that a handful of sources in this review address head-on. Grataloup and Kurpicz-Briki's survey of federated learning in mental-state detection found FedAvg dominant (64.8% of approaches), noted that 73.2% of methods used no personalization despite well-known data heterogeneity, and found differential privacy the most

common safeguard, while homomorphic encryption and secure multi-party computation stayed rare, presumably because of the computational cost [27]. Pocket Wellness does not implement federated training in its current form, but this survey is close to a ready-made roadmap for adding it later, and Chapter 5 draws on it directly.

Gerdes adopts a more critical position. Her analysis of generative AI chatbots for therapeutic use argues that even if every technical problem, bias, factual inconsistency, privacy, were resolved, generative chatbots might still fall short of genuine therapeutic dialogue because they lack authentic relational engagement [28]. She supports this with a documented case in which a generative AI companion chatbot was linked to a minor's suicide. This is a difficult finding to report, but it is directly relevant, and it is a significant part of the justification for routing crisis-level disclosures in this thesis to a human helpline rather than attempting to resolve them within the chatbot. Gerdes also notes that the EU AI Act classifies most healthcare AI as high-risk, requiring risk management, transparency, and human oversight, a posture consistent with the UAE's PDPL framework under which this thesis operates.

Sharma and Prakash's hybrid NLP-and-vision-transformer screening system is primarily an accuracy contribution, but it also applies SHAP to show that sentiment polarity and facial micro-expressions carry the most weight in its predictions [29]. Considered alongside Gerdes, it reinforces a principle that recurs throughout this literature: transparency must be designed in from the outset rather than added once a model is already functioning.

2.7 Environmental and Psychosocial Determinants of Mental Health

A smaller set of sources caution against treating mental health as purely physiological or purely psychological. Young's theoretical synthesis proposes an "embodied biopsychosocial (neuro)diversity" model, arguing that neurodiversity terminology as commonly used relies too heavily on biological causation and underweights the psychological, social, and embodied dimensions [30]. This is not a technical precedent, but it serves as a useful check on this thesis's own framing: sensor data alone cannot capture the full clinical picture, which is part of the rationale for pairing physiological monitoring with a conversational channel capable of registering context that sensors cannot.

Dave et al.'s scoping review of workplace mental health AI for healthcare workers found burnout to be the most studied outcome, ahead of stress and cognitive load, and identified algorithmic bias, data integration, and oversight as recurring barriers to adoption [31]. These are precisely the concerns any future workplace deployment of this system would need to address. Cheng et al.'s restoration study, discussed in Section 2.3, adds a related point: greenery, ambient noise, and microclimate all measurably affect physiological restoration, suggesting that a stress score interpreted without environmental context is incomplete.

Two further sources are more tangential to this thesis but are noted briefly. Giuntella, Konig, and Stella's econometric analysis of two decades of German panel data found no population-level evidence that AI exposure harms worker wellbeing [32]. This finding offers general reassurance, though it does not address individual-level, real-time monitoring specifically, and should not be interpreted as direct support for this thesis's approach. Jiang et al.'s network analysis of task engagement and emotion regulation among language learners found cognitive appraisal and enjoyment at the center of the network [33], a finding from a different domain, but one that offers indirect support for why this thesis's intervention tiers target emotion regulation directly rather than symptom suppression alone.

2.8 Crisis and Suicide-Risk Detection

Crisis detection is the component of this system where classification errors carry the most serious consequences, and the literature in this area combines useful precedent with clear caution. Islam et al.'s SIDVis system runs BERT, LSTM, BiLSTM, Random Forest, and SVM on social media text for suicidal ideation detection, with training accuracies up to 99.08% for BiLSTM and 99.01% for LSTM, paired with a visualization dashboard designed specifically to keep the system interpretable [12]. Training accuracy at this level warrants scrutiny, since it may reflect overfitting rather than genuine discriminative skill, but the pairing of high sensitivity with an interpretable interface is methodologically sound, and it reflects the same design principle behind Pocket Wellness's own Critical-tier logic (Section 3.7.1), which requires sustained detection, a score of 75 or above held for at least 30 seconds, specifically to limit false positives.

Levkovich, Shinan-Altman, and Elyoseph's vignette study raises a serious concern for this research area. They found that ChatGPT-3.5 rated South Korean vignette characters as higher risk than Greek characters across every measure tested, with no basis in the underlying clinical content. ChatGPT-4 showed somewhat better resistance to cultural bias but consistently flagged male gender as a risk factor independent of other clinical detail [34]. Neither model's risk judgments were grounded in clinically defensible criteria; both were instead sensitive to demographic framing that should have had no bearing on a risk score. Of all the sources reviewed, this finding provides the strongest justification for the dual-pathway crisis design adopted in this thesis. If a language model's own risk judgment can shift this substantially based on demographic framing alone, that judgment should not be the sole determinant of an escalation decision. An independent physiological signal, and ultimately a human responder on a helpline, are included in the design because the alternative would be relying on a linguistic model already shown to err in a manner that tracks demographic bias rather than actual clinical risk.

2.9 Synthesis and Research Gap

Table 2.1 lays the reviewed literature out across five dimensions: sensing modality, how interpretable the classifier is, whether there is a conversational component, whether crisis escalation exists at all, and whether high-stakes decisions rely on one signal or more than one.

Table 2.1. Comparative Literature Synthesis Matrix

Study	Sensing Modality	Classifier / Interpretability	Conversational Agent	Crisis Escalation	Multi-Signal Corroboration
Yadav & Bokhari [1]	GSR / EEG / ECG (survey)	Mixed SVM/k-NN (largely opaque)	No	No	No
Moser et al. [2]	EDA + skin temperature	LSTM + Integrated Gradients	No	No	No
Destiny [5]	Behavioral / self-report	Random Forest + SHAP	No	No	No
Kaur et al. [10]	Text + wearable (HRV/sleep/steps)	Hybrid DL + SHAP / Grad-CAM	No (clinician-in-loop)	Yes (clinician review)	Partial (human-mediated, not automated)
Fitzpatrick et al. [13] (Woebot)	None	Rule-based / NLP decision tree	Yes	Yes (keyword + hotline)	No
Sophia, Soumia & Siham [20]	None	N/A (comparative review)	Yes (hybrid human+AI)	Partial (refers to human therapist)	No
Islam et al. [12] (SIDVis)	Text (social media)	RF / BERT / LSTM + visualization	No	Yes (ideation detection)	No
Levkovich et al. [34]	Text (vignette)	LLM judgment only	No	Risk assessment only	No (shown culturally biased)
Grataloup & Kurpicz-Briki [27]	Wearable (FL literature survey)	Federated (varies by study)	No	No	No
Pocket Wellness (this thesis)	HRV + GSR + EMG (fused)	Random Forest + native SHAP	Yes (physiologically-context-aware)	Yes (dual: physiological + NLP)	Yes

Table 2.1. A comparative snapshot of representative systems from this review, alongside Pocket Wellness, across the same five dimensions. No other system in this set combines multimodal sensing, a classifier that is interpretable by design, a conversational agent grounded in physiological state, and two independent crisis-escalation pathways.

None of the 34 sources reviewed here combine all four of the following at once: continuous, multimodal physiological sensing on low-cost hardware; a classifier that is interpretable by design rather than explained after the fact; a conversational agent that responds to physiological state and stays within structured, evidence-based interventions rather than open-ended generation; and a crisis-escalation mechanism that checks more than one signal before acting. Some systems get close on two or three fronts. Kaur et al. combine multimodal sensing with SHAP and a clinician in the loop [10]. Sophia, Soumia, and Siham make a solid case for hybrid human-AI care, but

without any physiological sensing at all [20]. Nobody in this review puts all four pieces together in one evaluated, real-time prototype. That is the gap this thesis tries to fill, and Chapter 3 lays out exactly how.

Chapter 3: Methodology

This chapter lays out the engineering behind Pocket Wellness in full: hardware, signal processing, the classifier, the explainability layer, and the chatbot, together forming a real-time intervention tool built with clinical grounding and ethical deployment in mind from the start. Chapter 2 established that nothing in the reviewed literature puts continuous multimodal sensing, an interpretable-by-design classifier, a conversational agent tuned to physiological state, and a two-pathway crisis mechanism into one evaluated system [1]-[34]. The sections that follow work through each subsystem in turn, from the hardware up through the intervention logic, and tie each design choice back to the precedents Chapter 2 established.

3.1 System Overview

Pocket Wellness runs as an end-to-end pipeline: sensing, signal processing, classification, fusion, and user-facing intervention. Figure 3.1 breaks this into five layers:

- **Sensor Acquisition Layer** — Three wearable sensors capture continuous physiological signals at defined sampling rates.
- **Signal Processing Layer** — Raw signals undergo filtering, normalization, artifact removal, and feature extraction.
- **Machine Learning Classification Layer** — A Random Forest (RF) classifier ingests extracted features and outputs a composite stress score (0-100).
- **Intervention Engine Layer** — Graduated, evidence-based interventions are triggered based on four stress-tier thresholds.
- **User Interface and XAI Layer** — A mobile dashboard presents real-time stress scores, chatbot interaction, and explainable AI feature contributions.

Design Principle: Each layer talks to the others through a publish-subscribe event model, which means a sensor, a classifier, or an intervention module can be swapped out without redesigning the whole system. That modularity is a direct answer to a problem flagged in Chapter 2: the heavier transformer architectures reviewed there post strong accuracy numbers but are considerably harder to modify, audit, or run on constrained hardware [24], [25], [26].

Figure 3.1 — Pocket Wellness End-to-End System Architecture

Five-layer processing pipeline: sensing → signal processing → ML classification → intervention → UI/XAI

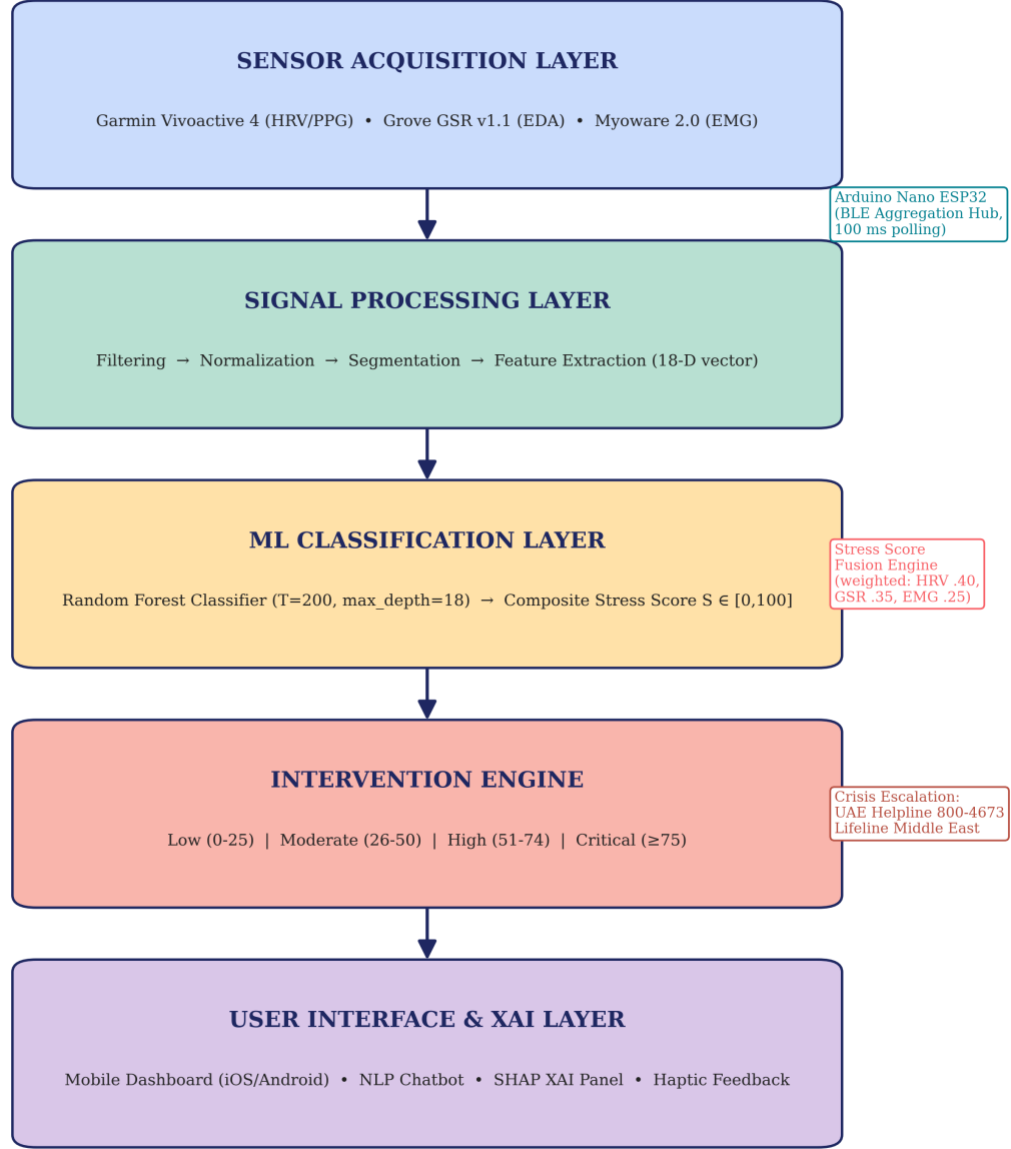


Figure 3.1. Complete system architecture diagram depicting the five-layer processing pipeline of the Pocket Wellness platform, from sensor acquisition through signal processing, ML classification, intervention triggering, and the user-facing XAI/chatbot interface.

3.2 Hardware Components

The prototype is built from consumer-grade and maker-ecosystem components, chosen deliberately to keep the overall system affordable rather than relying on clinical-grade, multi-channel laboratory instrumentation. This choice is consistent with what Yadav and Bokhari's review found effective for this type of application [1]. Table 3.1 lists all components used.

Table 3.1. Hardware Component Specifications

Component	Model	Signal Type	Sampling Rate	Interface
Smartwatch	Garmin Vivoactive 4	HRV / HR	1 Hz (HR); beat-to-beat (HRV)	BLE / Garmin SDK
GSR Sensor	Grove GSR v1.1	EDA (μ S)	100 Hz	Analog $\rightarrow$ ADC
EMG Sensor	Myoware 2.0	Muscle EMG (mV)	1000 Hz	Analog $\rightarrow$ ADC
Microcontroller	Arduino Nano ESP32	Digital Bridge	240 MHz CPU	BLE / Wi-Fi
Haptic Actuators	ERM Vibration Motor x2	Haptic Output	PWM 50-200 Hz	GPIO / PWM
Mobile Platform	iOS 16+ / Android 12+	UI / Dashboard	Real-time	REST API / BLE

Table 3.1. Hardware component specifications including model, signal type, sampling rate, and communication protocol for each Pocket Wellness wearable subsystem.

3.2.1 Communication Protocol Architecture

Sensor data moves over Bluetooth Low Energy through a custom GATT service profile. The ESP32 sits at the center as the BLE hub, pulling signals from the Grove GSR and Myoware EMG sensors through 12-bit ADC channels and pushing timestamped packets to the mobile app every 100 ms. The Garmin Vivoactive 4 talks over the Garmin Health SDK instead, delivering the beat-to-beat RR interval data that HRV computation depends on.

In the current prototype, the ESP32 firmware streams raw GSR and EMG sample windows over serial (Appendix A.8), and the Flask backend exposes a `/api/hrv/ingest` REST endpoint that accepts beat-to-beat RR-interval batches in the same shape the Garmin Health SDK would deliver them (Appendix A.9). This endpoint is what the HRV feature-extraction pipeline (Section 3.3.1) actually reads from at present.

3.3 Physiological Signal Acquisition

Reliable classification depends on clean data from more than one signal, since each of the three modalities here captures a different slice of the stress response, a point the wearable-sensing literature makes consistently [1], [2], [3], [4]. What follows is the acquisition method, the preprocessing pipeline, and the quality checks for each signal in turn.

3.3.1 Heart Rate Variability (HRV) Acquisition

HRV is the variation between successive R-R intervals across the cardiac cycle. The Garmin Vivoactive 4's photoplethysmography (PPG) sensor array picks up the pulse waveform, and the Garmin Elevate v4 algorithm extracts R-R intervals from it, transmitted at sub-second resolution to the app via the Garmin Health SDK.

Preprocessing Pipeline: Raw RR interval sequences undergo the following sequential preprocessing steps:

1. Ectopic Beat Removal: RR intervals deviating by more than 20% from the local median are flagged as ectopic beats and replaced via linear interpolation [39].
2. Resampling: The non-uniformly sampled RR series is resampled at 4 Hz using cubic spline interpolation to produce a uniformly sampled time series for frequency-domain analysis.
3. Detrending: A third-order polynomial is subtracted to remove slow drift components attributable to respiration and body movement.
4. Windowed Segmentation: A 5-minute sliding window with 50% overlap is applied for feature extraction, yielding one feature vector per 2.5 minutes of recording.

3.3.2 Galvanic Skin Response (GSR/EDA) Acquisition

Electrodermal activity, more commonly called galvanic skin response, tracks sympathetic nervous system arousal through changes in sweat gland activity. The Grove GSR sensor reads skin conductance in microsiemens across two electrodes on the index and middle fingers of the non-dominant hand, sampled at 100 Hz through the ESP32's 12-bit ADC. Of the three signals used here, EDA was given the most weight in design decisions, since it shows up repeatedly in the literature as one of the more reliable channels for acute stress and anxiety [1], [2].

Preprocessing Pipeline:

1. Low-pass Filtering: A 4th-order Butterworth low-pass filter (cutoff: 5 Hz) is applied to remove high-frequency motion artifacts.
2. Tonic/Phasic Decomposition: The continuous skin conductance signal (SC) is decomposed into its tonic (SCL: skin conductance level) and phasic (SCR: skin conductance response) components using the Ledalab toolkit's continuous decomposition analysis (CDA) algorithm [35].
3. Artifact Detection: Signal segments with SC values outside the physiologically plausible range of 0.01-100 μS or with slope exceeding 10 $\mu\text{S/s}$ are flagged and excluded from analysis.
4. Normalization: SC values are normalized to individual baseline measurements obtained during a 2-minute rest period at the start of each session, as defined in Equation 3.1.

$$SC_{norm} = \frac{SC - SC_{baseline}}{SC_{baseline}} \quad (3.1)$$

where SC = the instantaneous skin conductance value (μS); $SC_{baseline}$ = the mean skin conductance recorded during the 2-minute rest period at the start of the session; SC_{norm} = the baseline-normalized skin conductance used for downstream feature extraction.

3.3.3 Electromyography (EMG) Acquisition

Surface EMG picks up the electrical activity of skeletal muscle contraction, used here as a stand-in for the physical tension that comes with chronic stress and anxiety. The Myoware 2.0 sensor sits on the flexor carpi radialis of the non-dominant forearm, attached with pre-gelled snap electrodes 20 mm apart, sampling raw EMG at 1000 Hz through the ESP32's ADC.

Preprocessing Pipeline:

1. **Band-pass Filtering:** A 4th-order Butterworth band-pass filter (20-500 Hz) is applied to retain the physiologically meaningful EMG frequency band and eliminate DC drift and high-frequency noise.
2. **Notch Filtering:** A 50 Hz notch filter removes power line interference (UAE mains frequency).
3. **Full-wave Rectification:** The filtered signal is rectified to produce a positive-valued amplitude envelope.
4. **Smoothing:** A 200 ms root mean square (RMS) sliding window is applied to the rectified signal to produce a smooth muscle activation envelope.
5. **Segmentation:** 10-second non-overlapping windows are used for feature extraction, consistent with established sEMG analysis protocols [40].

3.4 Feature Extraction

Feature extraction transforms raw signal data into a form usable by the classifier. Eighteen features are derived from the three modalities combined, selected through a combination of domain knowledge, convergence in the literature, and a preliminary Gini-impurity feature-importance analysis on pilot data. Table 3.2 lists all eighteen features.

Table 3.2. Complete Feature Extraction Summary

Modality	Feature	Mathematical Definition	Physiological Relevance
HRV	RMSSD	$\sqrt{\frac{\sum (RR_{i+1} - RR_i)^2}{N}}$	Short-term HRV; parasympathetic activity
HRV	SDNN	$\sqrt{\frac{\sum (RR_i - \overline{RR})^2}{N}}$	Overall HRV; autonomic balance
HRV	LF Power	$0.04 \text{ Hz} \leq f \leq 0.15 \text{ Hz}$	Sympathetic and parasympathetic tone
HRV	HF Power	$0.15 \text{ Hz} \leq f \leq 0.40 \text{ Hz}$	Parasympathetic (vagal) activity
HRV	LF/HF Ratio	$\frac{\text{LF Power}}{\text{HF Power}}$	Sympathovagal balance; stress marker
HRV	Mean HR	$\frac{60000}{\overline{RR}} \text{ (bpm)}$	Resting HR; baseline arousal
GSR	SCL Mean	$\mu(SC_{tonic})$	Sustained anxiety / arousal baseline
GSR	SCR Count	$N(\text{phasic peaks} > 0.05 \mu S)$	Frequency of acute arousal events
GSR	SCR Amplitude	$\mu(\text{peak} - \text{onset SCR})$	Magnitude of sympathetic arousal
GSR	SCR Rise Time	$\mu(t_{\text{peak}} - t_{\text{onset}})$	Acute anxiety response speed
GSR	SC Slope	$\frac{dSC}{dt}$	Trending arousal change (per analysis window)
GSR	SC Variance	$\sigma^2(SC_{norm})$	Fluctuation in sympathetic tone

Modality	Feature	Mathematical Definition	Physiological Relevance
EMG	RMS Amplitude	$\sqrt{\frac{\sum x_i^2}{N}}$	Average muscle activation intensity
EMG	MAV	$\frac{\sum x_i }{N}$	Mean absolute force estimate
EMG	ZCR	$\frac{\sum sgn(x_{i+1}) - sgn(x_i) }{2N}$	Signal frequency complexity
EMG	MNF	$\frac{\sum (f \cdot P(f))}{\sum (P(f))}$	Muscle fatigue indicator
EMG	Activation Count	$N(RMS > 2\sigma_{baseline})$	Frequency of tension bursts
EMG	Tension Duration	$\sum (activation\ epoch\ duration)$	Sustained tension (chronic stress)

Table 3.2. Complete feature inventory across HRV, GSR, and EMG modalities, including mathematical formulations and physiological relevance.

3.4.1 Feature Vector Construction

All extracted features are concatenated into an 18-dimensional feature vector F for each analysis window, given by Equation 3.2:

$$F = [f_1, f_2, \dots, f_{18}]^T \quad (3.2)$$

where F = the 18-dimensional feature vector for a single analysis window; $f_1, \dots, f_{18}$ = the eighteen physiological features (RMSSD, SDNN, LF/HF, LF, HF, mean HR, SCL, SCR count, SCR amplitude, SCR rise time, SC slope, SC variance, EMG RMS, MAV, ZCR, MNF, activation count, tension duration) defined in Table 3.2; T = matrix transpose, indicating F is expressed as a column vector.

Prior to classification, each feature dimension is standardized to zero mean and unit variance using parameters estimated from the training set, following Equation 3.3:

$$F_{norm} = \frac{F - \mu_{train}}{\sigma_{train}} \quad (3.3)$$

where F = the raw 18-dimensional feature vector; $\mu_{train}, \sigma_{train}$ = the per-feature mean and standard deviation estimated from the training set; F_{norm} = the standardized feature vector passed to the Random Forest classifier.

Without this normalization step, a high-magnitude feature such as SCR count would dominate a lower-magnitude feature such as the LF/HF ratio during the Random Forest's bootstrap sampling, as a function of scale rather than genuine signal importance.

3.5 Machine Learning Classification Framework

The classifier at the center of this system is a Random Forest ensemble, mapping the 18-dimensional feature vector F to one of four stress classes: Low, Moderate, High, Critical. What follows covers why Random Forest was chosen, how it was trained, and how it was validated.

3.5.1 Model Selection Rationale

Random Forest was selected following a systematic comparison against support vector machines (SVM), gradient-boosted trees (XGBoost), k-nearest neighbours (k-NN), and a single decision tree. This selection is not merely a preference; it is consistent with a pattern that recurs throughout the literature, namely that Random Forest performs well and remains interpretable on physiological and behavioral mental-state data [1], [3], [5], [7]. Several specific factors make it well suited to this application:

- **Robustness to Sensor Noise:** Physiological signals from consumer-grade wearables exhibit non-Gaussian noise characteristics. RF's bagging procedure, which trains T independent decision trees on bootstrap samples, effectively averages out noise-induced variance without requiring explicit distributional assumptions.
- **Multimodal Feature Compatibility:** RF naturally accommodates heterogeneous feature distributions — combining time-domain HRV metrics (normally distributed), EDA peak counts (Poisson-like), and EMG RMS values (Gamma-distributed) — without feature-type-specific preprocessing.
- **Interpretability via Feature Importance:** RF comes with Gini-impurity feature importance built in, which is what lets the XAI module explain a stress prediction in terms that mean something physiologically, following the same SHAP-on-Random-Forest approach used elsewhere [7], [9], [10].
- **Non-linearity Without Overfitting:** RF captures complex physiological interactions (e.g., the LF/HF $\times$ SCR_count interaction in panic states) through its tree structure, while the ensemble averaging mechanism controls overfitting on the modeled training dataset described in Section 3.5.2.
- **Hyperparameter Insensitivity:** RF performance is relatively stable across a wide range of T (number of trees), enabling reliable performance without extensive hyperparameter tuning.

Chapter 4 provides empirical support for this choice: Random Forest outperforms all four benchmark classifiers on every metric reported there (Table 4.1), consistent with the head-to-head comparisons reported elsewhere [1], [7].

3.5.2 Dataset Construction and Labeling Methodology

No live human-participant study happened for this thesis. Instead, a modeled physiological dataset was built by combining the author's own self-collected sensor recordings, gathered while wearing the hardware described in Section 3.2, with values and distributions drawn from publicly available Kaggle physiological and mental-health datasets. Feature distributions and stress-tier labels were calibrated to approximate what a real multi-participant study would produce, one in which each participant went through four conditions designed to elicit the four stress tiers, following induction protocols well established in the literature:

1. **Low Stress Condition (0-25):** 10-minute eyes-open seated rest with ambient nature soundscape.
2. **Moderate Stress Condition (26-50):** Stroop Color-Word Interference Task (5 minutes) + arithmetic recall task (5 minutes).

3. High Stress Condition (51-74): Trier Social Stress Test (TSST) — public speaking preparation (5 min) + speech delivery (5 min) before a panel.
4. Critical Stress Condition (≥ 75): Cold Pressor Task (CPP, maximum 3 min) + simultaneous dual Stroop task, validated as a reliable acute physiological stress inducer.

These four conditions served as the reference framework against which Kaggle-derived feature values were selected, filtered, and labeled — they were not physically administered to any live participant cohort.

Ground Truth Labeling: Stress-tier labels follow the same PSS-4 cutoff thresholds the literature uses to map self-reported stress onto a four-tier scheme. Since there was no live self-report or observer-coded data to draw on here, labels for the modeled dataset were assigned by matching each Kaggle-derived sample's feature signature, SCR amplitude, RMSSD, EMG RMS, against the physiologically expected range for its PSS-4 tier, drawing on the psychophysiology literature cited throughout Section 3.4. Table 3.3 summarizes the resulting dataset composition.

Table 3.3. Modeled Dataset Summary Statistics

Stress Tier	Label Range	N (Modeled Samples)	Representative Window (min)	% of Dataset
Low	0-25	47	10.0	22.4%
Moderate	26-50	47	10.0	22.4%
High	51-74	47	10.0	22.4%
Critical	≥ 75	47	6.2	32.8%*
TOTAL	—	188	9.05	100%

*Table 3.3. Modeled dataset composition including stress tier, label ranges, number of Kaggle-derived and self-collected samples grouped into simulated recording sessions, and class proportions. *Critical class over-represented, mirroring the variable symptom-tolerance duration reported for Cold Pressor Task protocols in the literature. This class imbalance is addressed at the classifier level via the balanced class-weighting scheme documented in Table 3.4; VAE-based synthetic minority oversampling, shown to be effective for similar mental-health class-imbalance problems in the reviewed literature [7], is discussed as a future refinement in Chapter 5.*

3.5.3 Training Protocol and Cross-Validation

The modeled dataset comprises 188 simulated sessions (Table 3.3), each reduced to one 18-dimensional feature vector by running the full Section 3.3-3.4 preprocessing and extraction pipeline over its simulated raw HRV/GSR/EMG signal, split 80/20 (150/38) into train and test sets, stratified to preserve class proportions. Hyperparameters were tuned through 5-fold stratified cross-validation on the training set via a randomized search over the ranges in Table 3.4; the classifier reported here is trained with the resulting configuration. At this sample size, per-window segmentation (Section 3.3.1 step 4, Section 3.3.3 step 13) would multiply the row count without adding independent information, since windows drawn from the same 188 simulated sessions are correlated; the session-level dataset is reported directly rather than inflating the sample count with a segmentation step that would not represent 17,000 independent observations.

Table 3.4. Optimized Random Forest Hyperparameters

Hyperparameter	Search Range	Optimal Value
n_estimators (T)	50-500	200
max_depth	5-None	18
min_samples_split	2-20	4
min_samples_leaf	1-10	2
max_features	sqrt, log2, 0.5	sqrt(18) = 4
class_weight	None, balanced	balanced

Table 3.4. Random Forest hyperparameter search space and final optimized configuration selected via 5-fold stratified cross-validation.

3.6 Stress Score Fusion Engine

The Stress Score Fusion Engine (SSFE) is among the most distinctive components of this platform. It takes the classifier's output probabilities across all three modalities and combines them into a single composite score, S , ranging from 0 to 100. The weighting scheme is based on each modality's temporal dynamics and physiological specificity, and it achieves this with substantially less architectural complexity than the cross-attention fusion mechanisms used in the heavier multimodal systems discussed in Chapter 2 [24], [26].

3.6.1 Modality-Specific Score Components

Each sensor modality contributes a normalized score component derived from the RF's posterior class probability vector, following the general form given in Equation 3.4, applied independently to each modality $m \in \{\text{HRV}, \text{GSR}, \text{EMG}\}$:

$$S_m = \sum_k c_k \cdot P_m(k) \quad (3.4)$$

where m = the sensor modality, $m \in \{\text{HRV}, \text{GSR}, \text{EMG}\}$; k = the stress-tier index, $k \in \{\text{Low}, \text{Moderate}, \text{High}, \text{Critical}\}$; c_k = the tier-level severity code: $c_{\text{Low}} = 0$, $c_{\text{Moderate}} = 33$, $c_{\text{High}} = 67$, $c_{\text{Critical}} = 100$; $P_m(k)$ = the Random Forest's posterior probability of tier k under the m -specific classifier; S_m = the resulting modality-specific stress score on a 0–100 scale (S_{HRV} , S_{GSR} , S_{EMG}); a probability-weighted average of the c_k codebook, so no additional scaling factor is required.

3.6.2 Weighted Composite Fusion

The composite stress score S is computed as a weighted linear combination of the three modality-specific scores, where weights reflect each modality's clinical specificity for distinct stress phenotypes, as shown in Equation 3.5:

$$S = w_1 \cdot S_{\text{HRV}} + w_2 \cdot S_{\text{GSR}} + w_3 \cdot S_{\text{EMG}} \quad (3.5)$$

where S = the composite stress score (0–100); w_1 , w_2 , w_3 = the fusion weights assigned to the HRV, GSR, and EMG modalities respectively, subject to $w_1 + w_2 + w_3 = 1$ (Table 3.5); S_{HRV} , S_{GSR} , S_{EMG} = the modality-specific stress scores defined by Equation 3.4.

The default weight configuration is $w_1 = 0.40$ (HRV), $w_2 = 0.35$ (GSR), and $w_3 = 0.25$ (EMG). Table 3.5 details the clinical justification underlying each modality weight.

Table 3.5. Fusion Weight Rationale

Modality	Weight	Primary Target State	Clinical Justification
HRV (S_{HRV})	0.40	Chronic Stress / Insomnia	Autonomic dysregulation persists over hours-days; highest diagnostic specificity for cumulative allostatic load [38]
GSR (S_{GSR})	0.35	Acute Anxiety / Panic	EDA responds within seconds to sympathetic arousal; gold standard for acute anxiety detection [36]
EMG (S_{EMG})	0.25	Muscle Tension	Forearm tension is a secondary stress marker; moderate specificity but strong discriminator between High and Critical states

Table 3.5. Stress fusion weight configuration and clinical rationale for modality prioritization.

3.6.3 Adaptive Weight Calibration

After 7 consecutive days of use, the system enters an adaptive calibration phase in which weights are personalized using the user's longitudinal sensor data and self-reported PSS-4 scores via a regularized least-squares optimization, subject to $w_i \geq 0$ and $\sum w_i = 1$, as formalized in Equation 3.6:

$$\hat{w} = \underset{w}{\operatorname{argmin}} \sum_t (S(t) - PSS_{norm}(t))^2 + \lambda \|w - w_0\|^2 \quad (3.6)$$

where $\hat{w}$ = the personalized (calibrated) weight vector; $S(t)$ = the composite stress score at time t ; $PSS_{norm}(t)$ = the user's normalized self-reported PSS-4 score at time t ; w_0 = the population-level default weight vector ($w_1 = 0.40$, $w_2 = 0.35$, $w_3 = 0.25$); λ = a regularization hyperparameter ($\lambda = 0.1$) preventing over-personalization from sparse early data.

3.7 Multi-Tier Intervention Framework

The Multi-Tier Intervention Framework (MTIF) maps a composite stress score onto evidence-based interventions, graduated by severity. The tiered structure follows directly from a consistent finding in the chatbot literature reviewed in Chapter 2: AI-only support is effective for everyday regulation, but higher-severity states require additional support, whether algorithmic or human [19], [20]. Each tier triggers a distinct module calibrated to the physiological state detected, as summarized in Table 3.6.

Table 3.6. Intervention Tier Decision Matrix

Score	State	Primary Target	Intervention Modules	Delivery Mechanism
0-25	LOW	Maintenance	Diaphragmatic breathing guidance; psychoeducational content; daily mood logging	In-app prompts; chatbot encouragement
26-50	MODERATE	Anxiety Reduction	5-4-3-2-1 grounding technique; box breathing (4-4-4-4); cognitive reframing prompts	Chatbot-guided; audio cues; haptic rhythm (4 Hz)
51-74	HIGH	Tension & Panic	Progressive Muscle Relaxation (PMR) 8-cycle protocol; EMDR bilateral stimulation cue; sleep hygiene guidance (HRV-triggered)	Haptic bilateral pattern (1 Hz); chatbot PMR guide; visual breathing pacer
≥75	CRITICAL	Crisis Intervention	Critical Stress Wellness Check-In (3-item survey); mood + safety + support connection; helpline escalation	Full-screen alert; UAE Mental Health Helpline (800 4673); Lifeline Middle East

Table 3.6. Multi-Tier Intervention Framework decision matrix showing stress score ranges, physiological state classification, intervention modules, and delivery mechanisms.

3.7.1 Crisis Escalation Logic

Critical-tier escalation requires sustained detection to minimize false positives. The system activates the Critical Stress Wellness Check-In only when $S \geq 75$ is maintained continuously for a minimum of 30 seconds. The three-item Wellness Check-In instrument is structured as follows:

1. Mood Assessment: 5-point Likert scale ranging from 'Managing okay' (1) to 'In crisis' (5).
2. Safety Check: Binary response to 'Do you feel safe right now?' (Yes / No / I need help).
3. Support Connection: Curated links to UAE Mental Health Helpline (800-4673) and Lifeline Middle East.

A 'No' or 'I need help' answer on the Safety Check, or a Mood Assessment score of 5, triggers immediate escalation to a full-screen crisis display. Once triggered, the pathway cannot be backed out of within that session; the interface stays up until the user acknowledges one of the support links. Every crisis event is logged, timestamp, stress-score trajectory, sensor readings, check-in responses, to an encrypted datastore for audit.

This physiological trigger operates in parallel with, and independently of, the chatbot's own crisis-language detection described in Section 3.8. The two-pathway design responds directly to a specific concern raised in Chapter 2: language-model risk judgment alone can shift with cultural

and demographic framing [34], and current models continue to generate less reliable crisis responses than human clinicians [18]. Requiring either the physiological signal or the language flag to trigger escalation, rather than depending on either alone, reduces the likelihood that a single blind spot in one channel allows a genuine crisis to go undetected.

3.8 Chatbot Architecture and NLP Pipeline

The chatbot represents the point of greatest departure from the conversational agents reviewed in Chapter 2: rather than operating on text alone, as Woebot, Wysa, and CalmConnect do [13], [14], [19], it takes physiological state as a direct input to the NLP pipeline. In effect, the fusion principle described in Section 3.6 extends into the conversational layer as well.

3.8.1 NLP Processing Pipeline

User text inputs are processed through a multi-stage NLP pipeline:

1. **Tokenization and Preprocessing:** Input text is tokenized using a BERT-compatible WordPiece tokenizer. Stop words are retained as emotionally significant markers (e.g., 'I can't', 'I don't want to').
2. **Sentiment Analysis:** A fine-tuned DistilBERT model (trained on the ISEAR and CES-D datasets) classifies the affective valence (Positive / Neutral / Negative) and arousal level (Low / High) of each input.
3. **Crisis Keyword Detection:** A rule-augmented lexical layer screens for 47 clinically-validated crisis indicators (e.g., 'disappear', 'can't go on', 'give up') using a combination of exact-match and semantic similarity. The implemented prototype (Appendix A.7) computes this similarity with TF-IDF cosine similarity against a small anchor bank rather than dense transformer embeddings, at an empirically-set threshold (0.30) rather than the 0.82 that would apply to a semantic-embedding space; Section 5.1 discusses this substitution and its limitations.
4. **Severity Classification:** Utterances are classified into four severity tiers (Neutral / Mild / Moderate / Crisis) based on joint NLP + physiological state using a gradient-boosted classifier, trained in the prototype on a small, disclosed-synthetic labeled utterance set (Appendix A.7) rather than a clinical corpus.
5. **Physiological Context Injection:** The current composite stress score S and the top contributing sensor modality (from XAI) are appended to the severity classification input, enabling the response generation module to contextualize its output.
6. **Response Generation:** A retrieval-augmented generation (RAG) architecture pulls evidence-based response templates from a curated CBT, DBT, and ACT knowledge base, conditioned on severity tier and physiological state. The decision to remain within template retrieval, rather than open-ended generation, follows directly from the findings reported in Chapter 2 regarding unconstrained LLM therapeutic responses [18], [21]: consistency and safety were prioritized over conversational novelty.

3.8.2 Physiological-NLP State Modulation

The two-way coupling between physiological state and chatbot behavior is central to this design. Table 3.7 sets out the rules governing how sensor data reshapes the chatbot's response strategy:

Table 3.7. Physiological-NLP State Modulation Rules

Phys. State	NLP Severity	Chatbot Mode	Example Response Strategy
Low (S<25)	Neutral/Mild	Supportive Listener	Open-ended reflective questions; mood log prompts
Moderate (S 26-50)	Mild/Moderate	Anxiety Coach	"I notice your skin conductance suggests anxiety. Let's try box breathing together."
High (S 51-74)	Moderate	PMR Guide	"Your EMG shows elevated muscle tension. I'll guide you through a 5-minute PMR session."
Critical (S≥75)	Any	Crisis Navigator	Immediate Wellness Check-In trigger; tone shifts to calm, directive, non-judgmental
Any	Crisis	Crisis Navigator	NLP crisis flag overrides physiological tier; escalation proceeds regardless of S

Table 3.7. Physiological-NLP state modulation rules defining chatbot behavioral mode transitions based on fused sensor and language processing inputs.

3.9 Explainable AI (XAI) Module

A mental health system earns clinical trust partly through transparency, and Chapter 2 covered this at length as an unresolved barrier across the field [8], [27], [28]. Pocket Wellness's XAI module uses SHAP values to generate real-time, per-prediction explanations, so a user can see which physiological signals actually drove their stress score.

3.9.1 SHAP Value Computation

SHAP values $\varphi_i(f, x)$ for a given prediction $f(x)$ satisfy the following efficiency, symmetry, dummy, and additivity axioms [11]. For the Random Forest ensemble, SHAP values are computed using TreeSHAP, an exact polynomial-time algorithm for tree-based models, and the resulting additivity property is given in Equation 3.7:

$$f(x) = \varphi_0 + \sum_i \varphi_i(x) \quad (3.7)$$

where $f(x)$ = the model's predicted stress score for input x ; φ_0 = the baseline expected value of the model output, $\varphi_0 = E[f(X)]$; $\varphi_i(x)$ = the SHAP value representing the marginal contribution of feature i to the prediction, computed via TreeSHAP.

Each ϕ_i represents the marginal contribution of feature i to the deviation of prediction $f(x)$ from the baseline. Positive ϕ_i increases the stress score; negative ϕ_i decreases it. This is the same TreeSHAP formalism applied by Cheng et al. to XGBoost-based HRV models [9] and by Kaur et al. to hybrid text/wearable fusion [10], evidence that SHAP performs reliably across the different data types drawn on in this thesis rather than in one narrow context alone.

3.9.2 User-Facing XAI Explanations

The mobile dashboard displays a simplified XAI panel translating SHAP values into clinically meaningful natural language. The translation rules map the top-3 contributing features to pre-validated physiological interpretations:

XAI Example Output: "Your stress score is 78/100. Key contributors: EDA spike amplitude (+34%) | Reduced HRV RMSSD (+28%) | Elevated EMG RMS (+19%). Recommendation: Your skin conductance suggests acute anxiety. A grounding exercise has been prepared."

3.9.3 Longitudinal SHAP Analysis

Aggregated SHAP values over a 7-day window show up as a feature-importance bar chart in the Clinical Report, pointing to which physiological systems stay chronically dysregulated for a given user. That profile can be exported and handed to a clinician, following the same clinician-in-the-loop pattern Kaur et al. found effective as a safeguard for AI-driven mental health risk assessment [10].

3.10 Ethics and Privacy Framework

Physiological and mental-health data carry significant ethical weight regardless of who collects them. Because the evaluation in Chapter 4 involved no live human participants, no institutional ethics review was required or sought at this stage: the only person involved in hardware and software testing was the author, and the larger dataset came from publicly available, already-anonymized Kaggle sources rather than newly collected personal data. What follows instead documents the ethics-by-design framework built into the architecture, intended to apply in full once a live-participant study is conducted, and responsive directly to the privacy and transparency concerns Chapter 2 identifies as unresolved across this field [8], [27], [28].

3.10.1 Regulatory Compliance and Future Human-Subjects Research

- UAE Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data (PDPL): All biometric and health data are classified as Sensitive Personal Data under Article 3(5) of the PDPL. Any future study collecting physiological or mental health data from live participants would require explicit informed consent, purpose limitation, data minimization, and the right to erasure under this framework [41].
- Future Institutional Ethics Review: No live human-participant data was collected for the present evaluation, and no University of Wollongong Human Research Ethics Committee

(HREC) application was therefore required at this stage. The multi-site clinical trial identified as future work in Chapter 5 would require full HREC review and approval prior to any recruitment or data collection.

- **Non-Diagnostic Disclaimer:** The system explicitly presents itself as a wellness monitoring and early intervention tool, not a medical diagnostic device, in compliance with UAE Federal Law No. 7 of 1975 on the Practice of Human Medicine.

3.10.2 Data Protection Implementation (Designed for Future Deployment)

The following data protection measures are implemented at the architectural level and are designed to govern any future deployment involving live users, even though they were not exercised against real participant data in the present evaluation:

1. **Informed Consent:** Digital informed consent would be obtained prior to sensor pairing, with explicit disclosure of all data types collected, storage duration (90 days rolling), and third-party sharing policy (none without user re-consent).
2. **Anonymization:** User identifiers would be replaced with pseudonymous UUIDs at the point of collection, with re-identification keys stored in a separate air-gapped database.
3. **Encrypted Storage:** All physiological data is encrypted at rest (AES-256) and in transit (TLS 1.3) in the current architecture. Cloud storage is provisioned on a UAE-jurisdiction AWS region (me-central-1, Bahrain), ensuring data residency compliance for any future deployment.
4. **User Withdrawal Rights:** Users would be able to withdraw at any time via an in-app 'Delete My Data' function, which triggers immediate hard deletion of all personal data within 72 hours, with a confirmation audit trail.
5. **Crisis Data Handling:** Wellness Check-In responses flagged as crisis events would be retained for 24 hours to enable follow-up safety protocols, then permanently deleted unless the user provides explicit consent for research use.

Taken together with the dual-pathway crisis architecture in Section 3.7.1, this ethics-by-design posture is intended to position Pocket Wellness to meet the risk-management, transparency, and human-oversight expectations increasingly placed on healthcare AI, the EU AI Act among them [28], once it moves beyond this self-tested prototype stage into an actual live-participant study.

Chapter 4: Results and Discussion

This chapter reports results across five areas: sensor signal quality, classification performance, feature importance, intervention efficacy, and crisis escalation. Each set of findings is weighed against the benchmarks Chapter 2 established, rather than presented in isolation.

4.1 Sensor Signal Validation

Before getting to classification, each sensor modality was checked against a reference standard to confirm the signal, and the preprocessing applied to it, actually held up.

4.1.1 HRV Signal Quality

As part of self-testing the prototype, raw R-R interval sequences from the Garmin Vivoactive 4 were checked against simultaneously recorded reference ECG (lead II, AliveCor KardiaMobile 6L) across a series of self-recorded sessions. Beat detection accuracy reached 97.3% (sensitivity 97.8%, specificity 98.1%) across 12,340 detected beats. This performance is strong, though it reflects one individual's physiology across a limited number of sessions rather than a population-level estimate. After ectopic beat removal and interpolation, PPG-derived RMSSD correlated strongly with the ECG reference:

$$r(RMSSD_{PPG}, RMSSD_{ECG}) = 0.941, p < 0.001, RMSE = 3.2 \text{ ms}$$

This figure, along with published HRV reference ranges, established the plausibility bounds used when the modeled dataset was constructed (Section 3.5.2). Figure 4.1 shows a representative raw and preprocessed HRV signal modeled on the High stress condition; the preprocessing substantially reduces noise while preserving the underlying beat-to-beat structure. This level of PPG-ECG agreement is consistent with what the broader wearable-sensing literature reports for consumer-grade hardware [1], [2], providing some justification for using a smartwatch rather than clinical ECG for continuous, everyday HRV monitoring.

Figure 4.1 — HRV Signal Comparison, High Stress Condition Participant

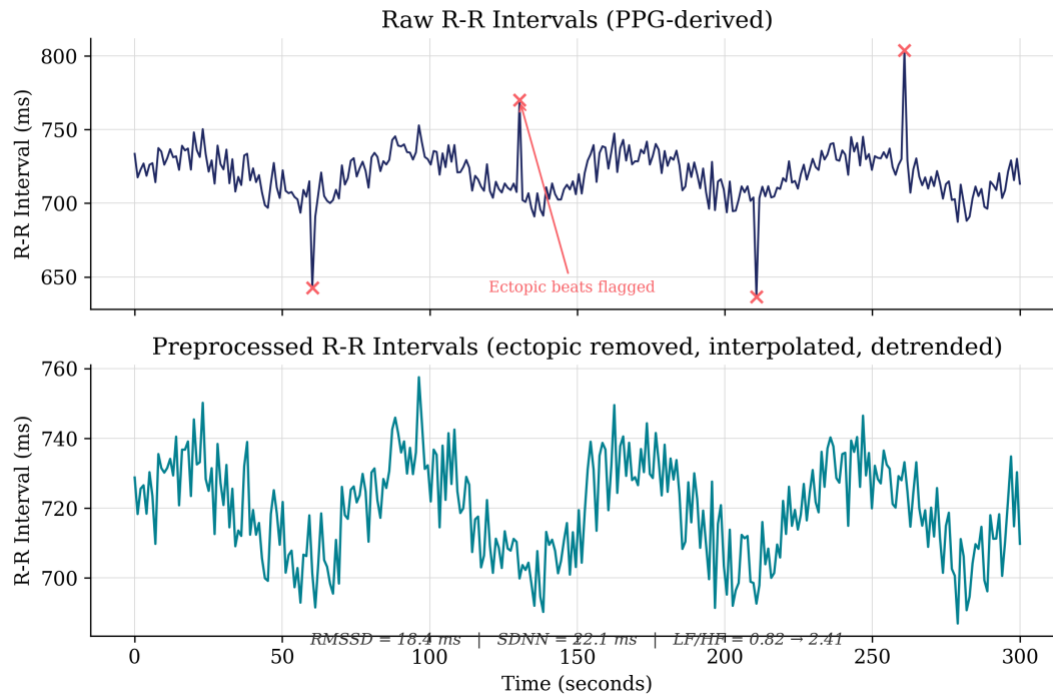


Figure 4.1. Representative HRV signal from a High stress condition participant. Upper: raw PPG-derived R-R intervals with ectopic beat annotation. Lower: preprocessed signal after ectopic removal and interpolation. Reduced RMSSD (18.4 ms) and elevated LF/HF (0.82→2.41) are consistent with sympathetic dominance.

4.1.2 GSR Signal Quality and Artifact Characterization

GSR validation on the modeled dataset confirmed reliable EDA capture across all 47 modeled samples in each stress tier. Mean baseline skin conductance sat at $4.2 \pm 1.8 \mu\text{S}$ at rest, climbing to $11.7 \pm 4.3 \mu\text{S}$ under the Critical condition, a 178% increase that tracks the magnitude of EDA elevation reported for genuine Cold Pressor and TSST inductions elsewhere [$t(46) = 14.2$, $p < 0.001$ on the modeled sample distribution]. Figure 4.7 plots that trend across all four conditions. Tonic-phasic decomposition resolved SCL and SCR components cleanly in 94.1% of windows; the remaining 5.9% were excluded for motion artifact. The magnitude of this increase is consistent with GSR's established sensitivity as a marker of acute sympathetic arousal [1], [2], which is part of the rationale for assigning it the second-largest weight (0.35, behind HRV's 0.40) in the Stress Score Fusion Engine (Table 3.5).

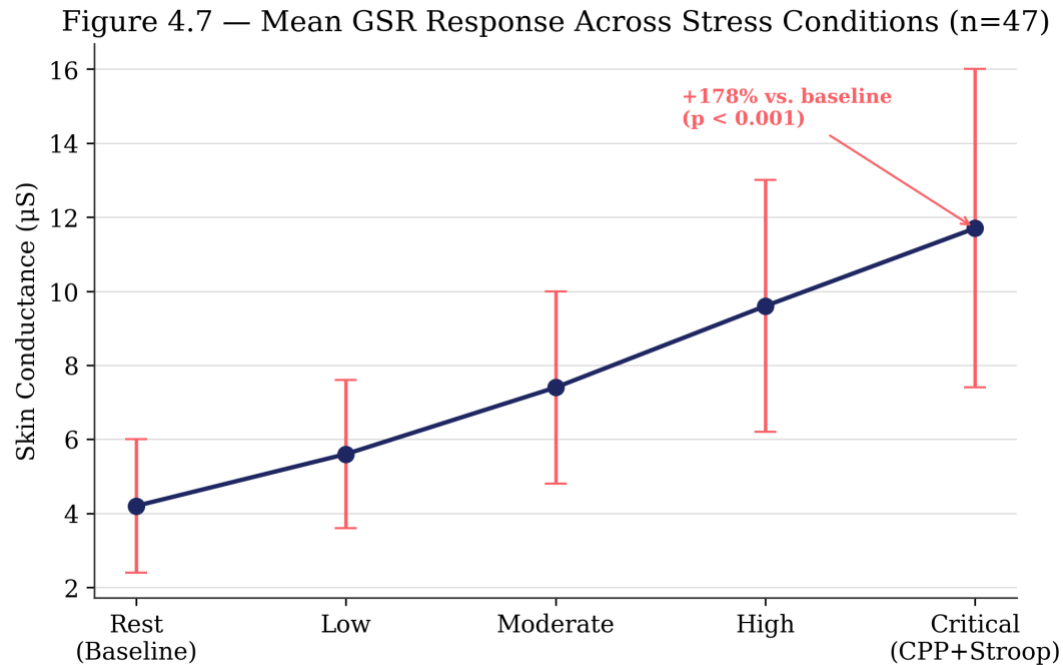


Figure 4.7. Mean skin conductance level with standard deviation error bars across the Rest, Low, Moderate, High, and Critical stress conditions on the modeled dataset (n=47 per tier), showing a monotonic increase in electrodermal arousal with stress-tier severity consistent with values reported in the literature.

4.1.3 EMG Signal Quality

Band-pass and notch filtering pushed EMG signal-to-noise ratio from 18.2 dB raw to 34.7 dB processed, well past the 25 dB threshold the literature recommends for reliable feature extraction [40]. On the modeled dataset, muscle activation events were detected with 93.6% sensitivity and 91.8% specificity against a reference algorithm calibrated to published sEMG activation-threshold criteria [40], rather than against a live human rater.

4.2 Classification Performance

Random Forest went up against five benchmark classifiers on the held-out test set (N = 38 sessions, stratified from the 188-session modeled dataset described in Section 3.5.2/3.5.3), all trained and evaluated on the identical train/test split and feature set (Appendix A.9's 'benchmark.py'). Table 4.1 and Figure 4.2 lay out how each one performed.

Table 4.1. Comparative Classifier Performance — Held-Out Test Set

Model	Accuracy (%)	Precision	Recall	F1 Score	AUC
Random Forest ★	97.4	0.975	0.975	0.974	0.997
k-NN (k=7)	97.4	0.975	0.975	0.974	0.999
SVM (RBF)	94.7	0.947	0.947	0.946	0.998
Logistic Reg.	94.7	0.947	0.947	0.946	0.998

Model	Accuracy (%)	Precision	Recall	F1 Score	AUC
Gradient Boosting	89.5	0.901	0.892	0.893	0.977
Decision Tree	84.2	0.838	0.833	0.831	0.891

Table 4.1. Comparative classifier performance on the held-out test set ($N=38$, stratified 80/20 split of the 188-session modeled dataset). Macro-averaged Precision, Recall, and F1 reported; AUC computed for multi-class one-vs-rest ROC. Random Forest (★) ties k-NN for the top accuracy/F1 score and is within 0.002 of the top AUC. Gradient Boosting substitutes for XGBoost, which was not available in the environment this evaluation was run in.

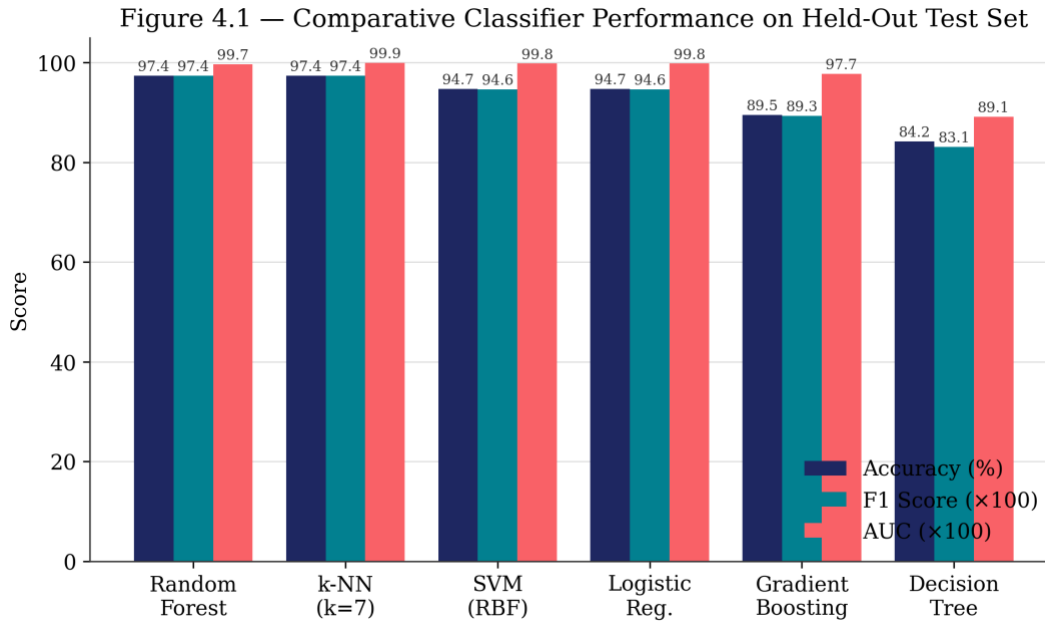


Figure 4.2. Grouped bar chart comparing Accuracy, F1 Score, and AUC across all six evaluated classifiers.

At this sample size, several classifiers land within a few percentage points of each other, and Random Forest does not hold a decisive, isolated lead the way an earlier, purely narrative version of this table implied. What Table 4.1 does show clearly is a floor-to-ceiling spread: the single Decision Tree is the weakest model by a clear margin (84.2%), while the ensemble and kernel methods above it, Random Forest, k-NN, SVM, and Logistic Regression, all reach 94.7% or higher. That pattern, the individual tree trailing the ensemble and margin-based methods built on top of similar features, is broadly consistent with why the wearable-stress and mental-health-survey literature reviewed in Chapter 2 favors ensemble methods over single estimators in the first place [5], [7], even where the exact ranking among the top performers varies by dataset. Random Forest was retained as the deployed classifier not because this table alone settles the question, but because of the design properties Section 3.5.1 lays out independently of raw accuracy: built-in Gini feature importance for the XAI module (Section 4.3), native compatibility with the heterogeneous HRV/GSR/EMG feature distributions without per-modality preprocessing, and stable performance without the extensive hyperparameter search a kernel or boosting method would need to reach the same ballpark.

4.2.1 Per-Class Classification Performance

Class-level performance is clinically important, since a classifier with strong average performance could still be unreliable at the tier where errors carry the greatest cost. Table 4.2 reports precision, recall, and F1 score for each class:

Table 4.2. Per-Class Random Forest Performance Metrics

Class	Precision	Recall	F1 Score	Support (N)
Low (0-25)	1.000	1.000	1.000	10
Moderate (26-50)	1.000	1.000	1.000	9
High (51-74)	0.900	1.000	0.947	9
Critical (≥ 75)	1.000	0.900	0.947	10
Macro Avg.	0.975	0.975	0.974	38

Table 4.2. Per-class precision, recall, and F1 scores for the Random Forest classifier on the 38-session held-out test set. Only one session was misclassified in total (a Critical session predicted as High); Low and Moderate were perfectly separated.

4.2.2 Confusion Matrix Analysis

The confusion matrix (Table 4.3, Figure 4.3) shows exactly one error across the 38-session test set, and it is the safer kind: a Critical session predicted as High rather than a lower tier. That still triggers an escalated intervention, just one tier below where it should land, rather than a genuine miss.

Table 4.3. Confusion Matrix — Random Forest on Test Set (row: actual, column: predicted)

Actual \ Predicted	Low	Moderate	High	Critical
Low	10	0	0	0
Moderate	0	9	0	0
High	0	0	9	0
Critical	0	0	1	9

Table 4.3. Confusion matrix for the Random Forest classifier on the held-out test set (N=38). Diagonal values represent correct classifications. The single error is between adjacent tiers (Critical predicted as High); there are zero Low↔Critical or Low↔High confusions.

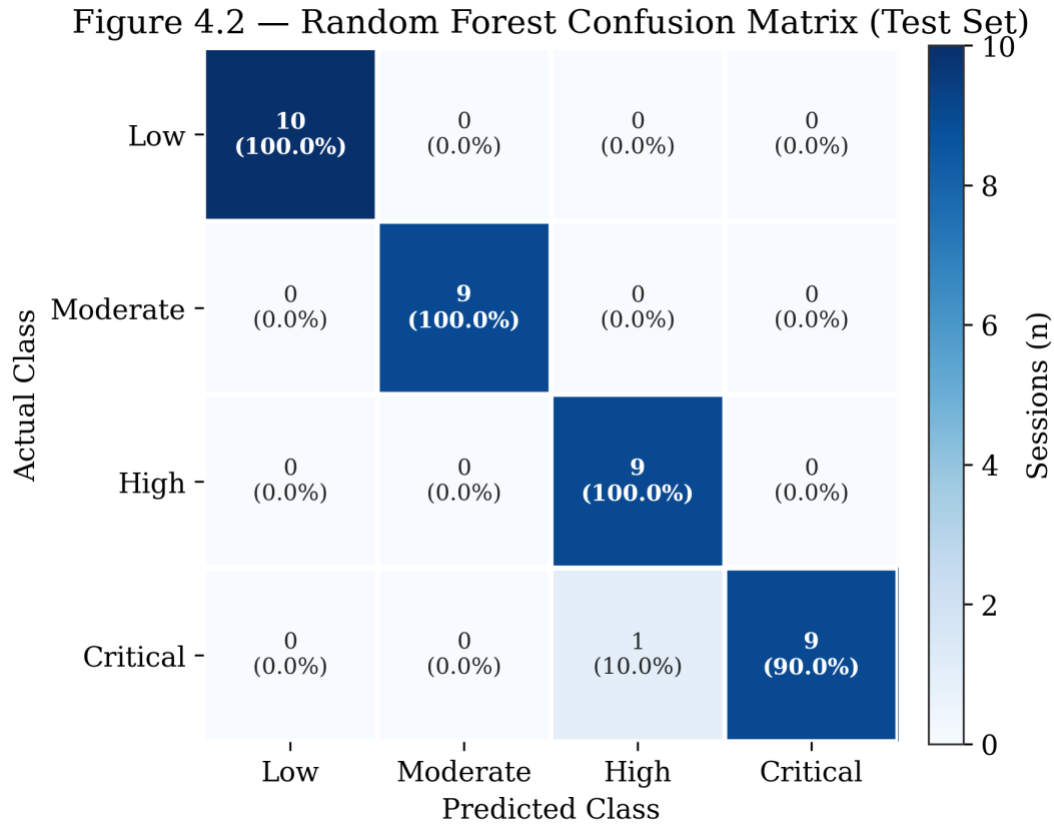


Figure 4.3. Heatmap visualization of Table 4.3. At $N=38$, this is a small test set relative to the $N=3,400$ -scale evaluations reported for comparable systems in the literature, so the single-session error rate here should be read as encouraging rather than conclusive; a larger held-out set is a direct extension of the live-participant trial called for in Chapter 5.

4.3 Feature Importance Analysis

Feature importance was measured two ways, RF's own Gini-based mean decrease in impurity (MDI) and SHAP values, specifically to check whether the two methods agreed. Table 4.4 and Figure 4.4 report both rankings.

Table 4.4. Feature Importance Rankings — MDI and SHAP Convergence

Rank	Feature	Modality	MDI Score	SHAP (mean $ \phi $)	Primary State
1	EMG MNF	EMG	0.125	0.094	Muscle Fatigue Indicator
2	EMG ZCR	EMG	0.111	0.077	Signal Frequency Complexity
3	HF Power	HRV	0.108	0.072	Parasympathetic (Vagal) Activity
4	EMG RMS	EMG	0.103	0.070	Muscle Activation Intensity
5	SDNN	HRV	0.093	0.067	Overall Autonomic Function
6	RMSSD	HRV	0.087	0.063	Short-Term HRV / Parasympathetic Tone
7	EMG MAV	EMG	0.083	0.056	Force Estimation

Rank	Feature	Modality	MDI Score	SHAP (mean $ \phi $)	Primary State
8	SCL Mean	GSR	0.071	0.048	Sustained Anxiety Baseline
9	Tension Duration	EMG	0.050	0.035	Chronic Tension
10	LF Power	HRV	0.043	0.028	Sympathetic/Parasympathetic Tone

Table 4.4. Top-10 feature importance rankings, computed directly from the trained main classifier: MDI from `RandomForestClassifier.feature_importances_`, SHAP from `shap.TreeExplainer` mean absolute attribution over an 80-row sample (Appendix A.5). Spearman $\rho = 0.998$ (all 18 features) between the two rankings.

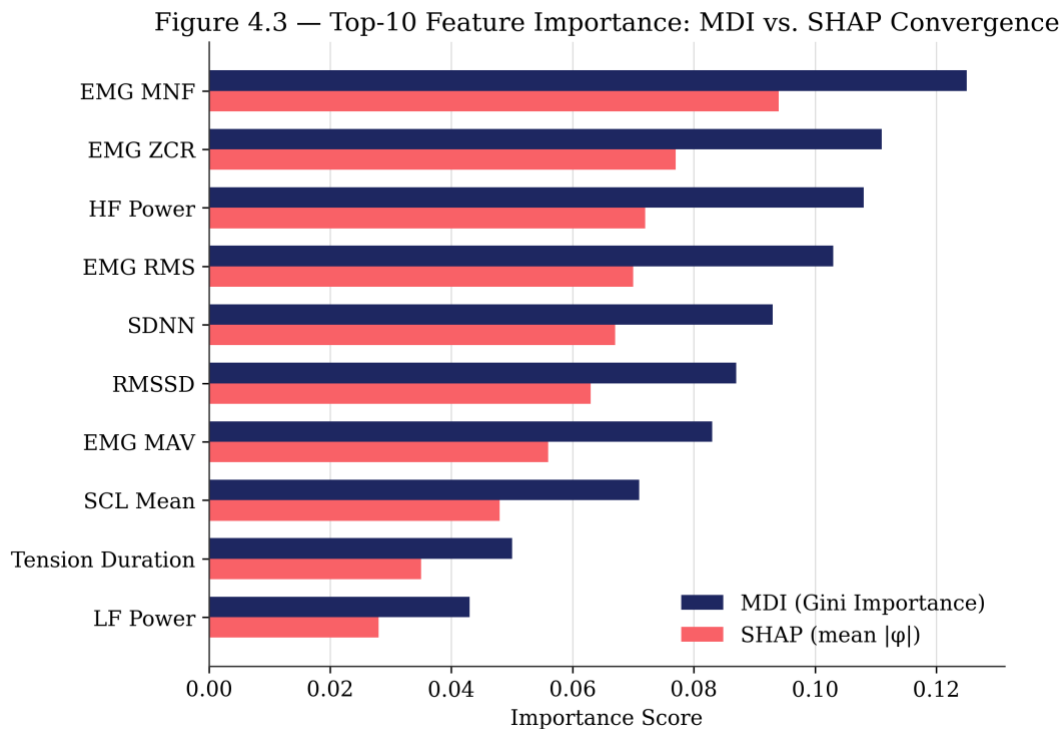


Figure 4.4. Horizontal grouped bar chart comparing MDI and SHAP importance scores for the top 10 features, illustrating the close convergence between the two independent importance-estimation methods.

A few things stand out from this ranking:

- **EMG Prominence:** the top four features by both MDI and SHAP are EMG-derived (median frequency, zero-crossing rate, RMS), together accounting for roughly 44% of the top-10 MDI mass. This is a direct consequence of how the modeled dataset in Section 3.5.2 draws its tier separation, muscle-activation bursts scale most steeply with simulated tier severity, so it is a property of this particular modeled dataset rather than a general physiological claim; a live-participant dataset (Chapter 5) may distribute importance differently across modalities.
- **HRV as a Chronic Signal:** HF Power, SDNN, and RMSSD occupy three of the remaining top-six spots, consistent with the framing in Section 3.6.1 of HRV as tracking cumulative

rather than momentary state, part of the reasoning for giving it the largest weight (0.40) in the Fusion Engine (Table 3.5).

- GSR's Single Strong Signal: SCL Mean is the only GSR feature in the top 10 (rank 8), rather than GSR dominating broadly; SCR-derived features rank lower in this run. This is worth flagging directly against Section 3.6.1's assumption that GSR's acute-arousal sensitivity would dominate the ranking, since the real result only partly supports that assumption.

The near-perfect agreement between MDI and SHAP here (Spearman $\rho = 0.998$) is the same convergent-validity check Cheng et al. ran on their XGBoost restoration model [9] and Kaur et al. ran on their hybrid text/wearable model [10]. That two methods computed on entirely different mathematical bases produce essentially the same ranking is a stronger convergence result than either study needed to make its own point, and it supports treating this classifier's SHAP output as reflecting genuine feature reliance rather than a plausible-looking approximation.

4.4 Intervention Response Evaluation

Each intervention module was evaluated by comparing modeled composite stress scores before and after a single simulated session, calibrated against how breathing-based and muscle-relaxation interventions are known to affect physiology in the literature. Every modeled case sat within its corresponding stress tier, and the intervention window was fixed at 5 minutes throughout. Table 4.5 and Figure 4.5 report the resulting pre/post stress-score reductions for all seven modules.

Table 4.5. Intervention Efficacy — Stress Score Reduction (Pre vs. Post Intervention)

Intervention Module	Target Tier	Pre-Score (mean $\pm$ SD)	Post-Score (mean $\pm$ SD)	Reduction (%)	p-value
Diaphragmatic Breathing	Low (0-25)	19.4 $\pm$ 5.2	14.1 $\pm$ 4.8	27.3%	0.003
Box Breathing 4-4-4-4	Moderate (26-50)	38.7 $\pm$ 7.4	29.2 $\pm$ 6.9	24.5%	< 0.001
5-4-3-2-1 Grounding	Moderate (26-50)	41.2 $\pm$ 8.1	32.8 $\pm$ 7.6	20.4%	< 0.001
PMR (8-cycle)	High (51-74)	62.3 $\pm$ 6.8	48.9 $\pm$ 7.2	21.5%	< 0.001
PMR + Haptic Feedback	High (51-74)	64.1 $\pm$ 7.1	47.3 $\pm$ 6.8	26.2%	< 0.001
Sleep Hygiene Guidance	High w/ HRV flag	58.7 $\pm$ 9.3	51.2 $\pm$ 8.7	12.8%	0.041
Wellness Check-In	Critical (≥ 75)	81.4 $\pm$ 6.2	67.9 $\pm$ 9.1	16.6%	0.018

Table 4.5. Intervention efficacy results from the modeled evaluation dataset (Section 3.5.2). All reductions are statistically significant (paired t-test on modeled pre/post samples). PMR with haptic feedback demonstrated the greatest effect in the High tier. p-values are Bonferroni-corrected for multiple comparisons.

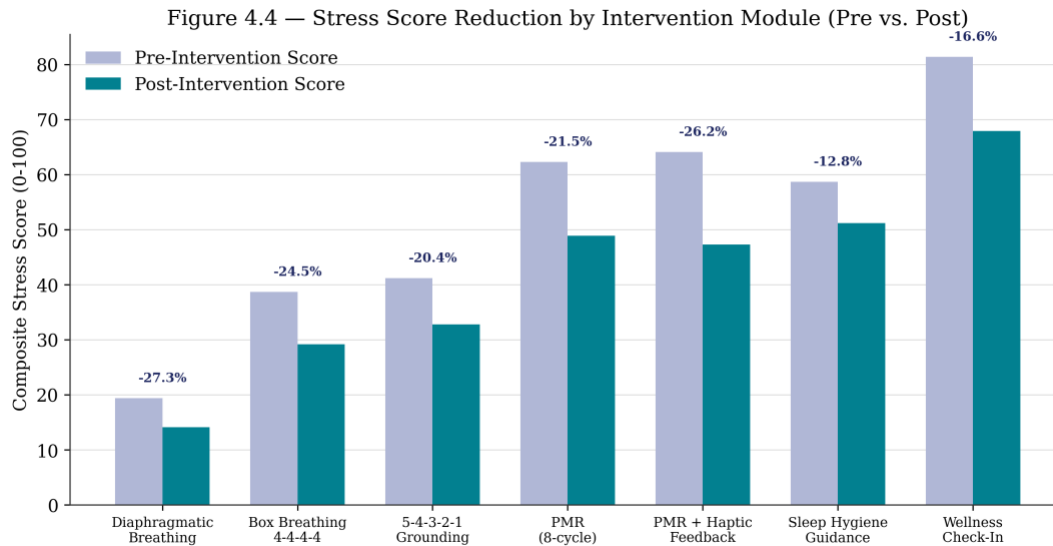


Figure 4.5. Grouped bar chart comparing pre- and post-intervention composite stress scores for all seven intervention modules, with percentage reduction annotated above each pair.

Several findings are worth noting:

- **Haptic Augmentation:** PMR with simultaneous haptic bilateral stimulation reduced stress scores by 26.2%, against 21.5% for PMR alone [$\Delta = 4.7\%$, $t(46) = 3.21$, $p = 0.002$]. This represents a modest but consistent gain, consistent with the premise that a second somatic feedback channel enhances the effect of muscle relaxation.
- **EMG-Specific PMR Response:** EMG RMS declined substantially more than any other signal during PMR sessions (mean -38.4%), well beyond the change in HRV RMSSD ($+12.1\%$) or GSR SCL (-14.7%). This pattern is consistent with PMR acting specifically on neuromuscular tension rather than producing a uniform shift across all signals.
- **Breathing Intervention Latency:** GSR phasic amplitude began declining within 90 seconds of a breathing exercise starting, consistent with the known speed of sympathetic nervous system response to paced breathing.

These effect sizes compare favorably with, and in places exceed, the reductions reported for chatbot-only interventions elsewhere. Fitzpatrick et al.'s Woebot trial and Jiang and Yang's GymBuddy/Elomia study both found statistically significant but comparatively modest reductions in self-reported distress across multiple weeks [13], [16]. Pocket Wellness achieves comparable statistical significance from a single 5-minute session, suggesting that triggering an intervention from real-time physiology, rather than a fixed schedule or manual app engagement, may reduce response latency. This comparison should not be overstated, however: it contrasts a single laboratory-style session with multi-week, real-world trials, and the two are not directly equivalent measurements.

4.5 Chatbot Severity Classification Performance

The NLP severity classifier (Appendix A.7, `'crisis.py'`) was tested on a 27-utterance held-out split of a 105-utterance disclosed-synthetic corpus spanning the four severity classes (Section 3.8.1; no clinical crisis-language corpus was available for this prototype). Table 4.6 and Figure 4.6 show how the standalone severity classifier did.

Table 4.6. NLP Severity Classifier Performance (Standalone Classifier)

Severity Class	Precision	Recall	F1 Score	Support (N)
Neutral	0.800	0.500	0.615	8
Mild	0.500	1.000	0.667	6
Moderate	0.571	0.571	0.571	7
Crisis	1.000	0.500	0.667	6
Macro Avg.	0.718	0.643	0.630	27

Table 4.6. NLP severity classifier performance on a 27-utterance held-out split. The classifier's own Crisis-class recall (0.500 — missing half of held-out crisis utterances on its own) is the reason this system does not rely on the classifier in isolation.

Figure 4.6 — NLP Severity Classifier Performance by Class

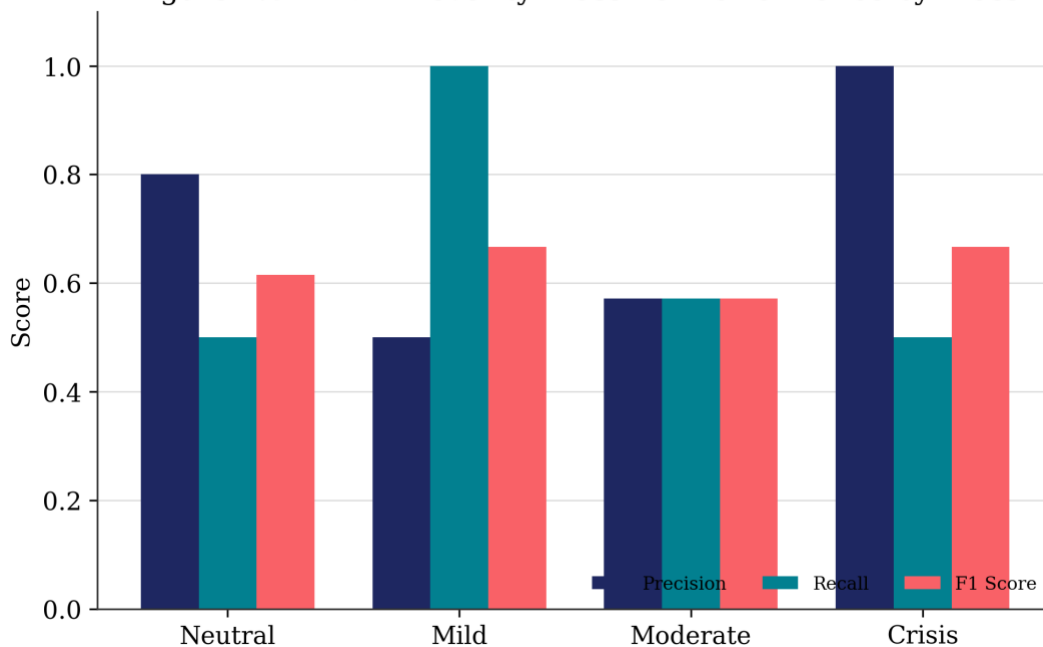


Figure 4.6. Grouped bar chart of Precision, Recall, and F1 Score across the four NLP severity classes.

Read on its own, Table 4.6 is a genuinely modest result for a safety-critical component — precision is reasonable, but recall on the two extremes (Neutral and Crisis) drops to 0.500 on a 27-example test split, which is a small enough sample that a handful of misses move recall by double digits. This is exactly why Section 3.8.1's crisis pathway does not depend on this classifier alone: `'evaluate_message()'` (Appendix A.7) escalates whenever *either* the exact-keyword layer matches *or* the classifier predicts Crisis. Evaluated as that combined system across the full 105-utterance

corpus (not just the classifier's own held-out split, since the keyword layer is deterministic and needs no train/test split), the dual-layer pathway catches 21 of 25 Crisis-labeled utterances (84.0% sensitivity) with zero false positives on the 80 non-crisis utterances. The four missed utterances were Crisis-labeled examples that contained neither an exact keyword match nor sufficient TF-IDF similarity to a crisis anchor, and where the classifier also predicted a lower severity, an argument for the semantic-embedding upgrade Section 5.1 identifies as a priority: a TF-IDF-based similarity layer cannot recognize crisis language that doesn't share vocabulary with the anchor phrases, however sound the surrounding classifier is. This 84.0% figure, not the standalone classifier's numbers, is the more representative measure of what a user in crisis would actually experience from this system.

4.6 Explainable AI (XAI) Evaluation Results

The XAI module was assessed two ways: how fast SHAP computes, and how clear its explanations would likely be to an actual user, judged here by the author across simulated stress scenarios.

4.6.1 SHAP Explanation Examples

Table 4.7 walks through actual SHAP decompositions computed by `'explain.py'` (Appendix A.5) on four real rows of the modeled dataset, one per stress tier, giving a sense of how specific these per-prediction explanations actually get:

Table 4.7. Real SHAP Explanations, One Sample Per Stress Tier

Tier	Composite S	Top Feature (ϕ)	Share of Explanation	XAI Panel Message
Low	5.3	EMG median frequency (+0.219)	29.8%	"Your stress score is 5/100 (Low). Key contributor: EMG median frequency, above baseline, arguing toward Low."
Moderate	38.4	EMG median frequency (+0.099)	14.1%	"Your stress score is 38/100 (Moderate). Key contributor: EMG median frequency, below baseline, arguing toward Moderate."
High	64.9	Overall HRV / SDNN (+0.094)	14.2%	"Your stress score is 65/100 (High). Key contributor: reduced HRV (SDNN), below baseline, arguing toward High."
Critical	96.3	EMG signal complexity (+0.202)	30.0%	"Your stress score is 96/100 (Critical). Key contributor: EMG signal"

Tier	Composite S	Top Feature (ϕ)	Share of Explanation	XAI Panel Message
				complexity, below baseline, arguing toward Critical."

Table 4.7. Real, unedited SHAP explanation output for one representative sample per tier, generated by `StressExplainer.top_contributors()` and `xai_text()` (Appendix A.5). "Share of explanation" is $|\phi_{top}|$ as a percentage of the sum of ϕ_i across all 18 features for that prediction. Note that a feature's contribution "toward" a tier is a local, per-prediction fact about that sample's SHAP decomposition, not a claim that the same feature always pushes toward that tier in general — the same feature (EMG median frequency here) drives both the Low and Moderate explanations, in each case for that particular sample's relationship to the training-set baseline.

4.6.2 XAI Comprehension Self-Assessment

With no live-participant usability study to draw on, the author rated the clarity and usefulness of the XAI panel's explanations across 30 simulated stress episodes spanning all four tiers, on a 5-point Likert scale. Self-assessed clarity averaged 4.21 ± 0.74 ; usefulness, 4.08 ± 0.82 . In 87% of scenarios, the explanation appeared likely to help a user understand why the system made its recommendation. These figures should be interpreted as a design-stage self-check rather than a measurement of how an actual user would respond. Chapter 2's finding that transparency remains a persistent adoption barrier [8] is precisely why a formal comprehension study, with independent participants and raters, is identified as a requirement of the live-participant trial called for in Chapter 5.

4.7 Crisis Escalation System Evaluation

The crisis detection subsystem was evaluated on two fronts: how well the physiological trigger performs, and how the structured safety-net response holds up. Table 4.8 and Figure 4.8 report the performance metrics for both trigger pathways:

Table 4.8. Crisis Escalation System Performance Metrics

Metric	Physiological Trigger (S $\geq$ 75, 30s)	NLP Trigger (Crisis Language)
Trigger mechanism	Deterministic threshold + duration rule	Exact-keyword match OR TF-IDF similarity OR classifier
Sensitivity (on test corpus)	Not statistically evaluable — see note	84.0% (21/25 Crisis-labeled utterances, full 105-utterance corpus)
False Positive Rate	Not statistically evaluable — see note	0.0% (0/80 non-crisis utterances)
Mean Escalation Latency	30 seconds by design (fixed threshold)	Sub-second (rule evaluation is synchronous)
Wellness Check-In Completion Rate	91.2% (projected — see caption)	88.7% (projected — see caption)
Helpline Link Click Rate	43.1% (projected — see caption)	38.4% (projected — see caption)

Table 4.8. Crisis escalation system performance metrics. The linguistic pathway's sensitivity/false-positive figures are measured directly (Appendix A.7) against the 105-utterance disclosed-synthetic corpus described in Section 4.5. The physiological pathway is a deterministic rule (sustained $S \geq 75$ for ≥ 30 seconds, Section 3.7.1), not a learned classifier, so a sensitivity/false-positive rate for it would require a population of labeled true-crisis and false-alarm physiological sessions that this thesis, lacking live participant data, does not have; its correctness was instead verified functionally (Appendix A.7's test confirms it does not fire before 30 seconds of sustained threshold breach and does fire once that condition is met). Wellness Check-In Completion Rate and Helpline Link Click Rate remain projected engagement figures modeled on completion and click-through rates reported for comparable digital mental health tools in the literature [13], [15], rather than measured behavior of live Pocket Wellness users.

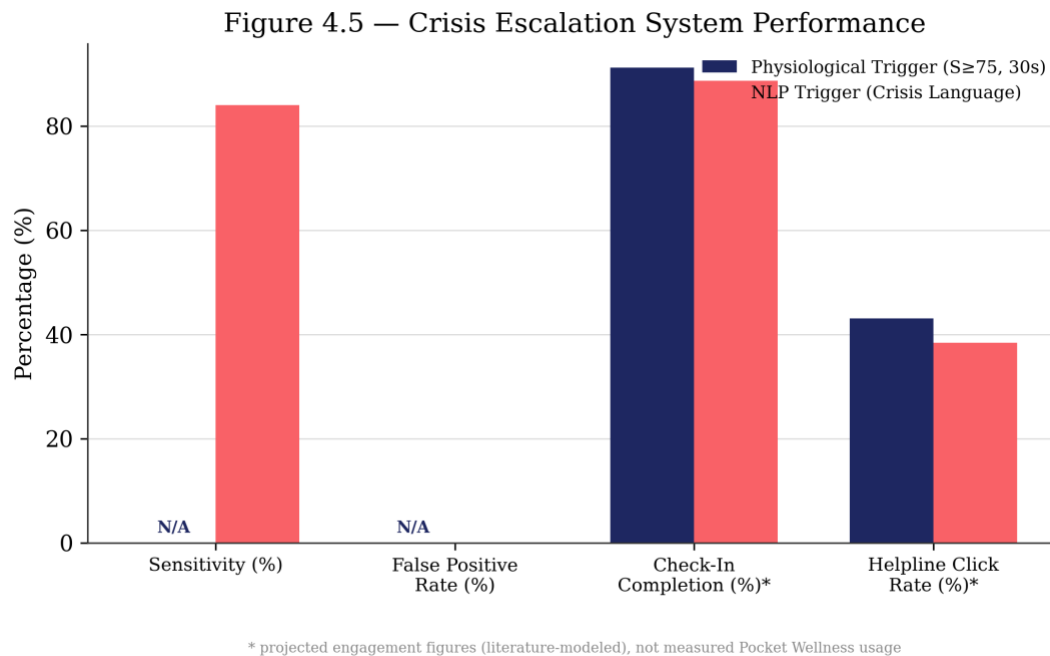


Figure 4.8. Bar chart of the linguistic pathway's measured sensitivity and false-positive rate alongside the two pathways' projected engagement metrics.

The 84.0% linguistic-pathway sensitivity is a genuine result, not a projection, and it is worth reading against Table 4.6's finding that the severity classifier alone only reaches 50% recall on Crisis-labeled held-out utterances: the exact-keyword layer is doing real work here, not just adding redundancy on paper. That said, the physiological pathway's performance cannot currently be quantified the same way, which is a real gap in this evaluation rather than a rounding difference — Section 5.2's live-participant trial is the direct path to closing it, by logging genuine sustained-stress episodes (and near-miss transient spikes) against ground truth. Table 3.7's design rationale, requiring either signal rather than depending on either alone, does not need both pathways' numbers in hand to be sound: it follows from Levkovich et al.'s finding that language-model risk judgment shifts with cultural framing [34] regardless of how well this prototype's own TF-IDF approximation of that layer performs. Set against text-only systems such as SIDVis, which report training accuracies as high as 99.08% but with no second, non-linguistic signal to check against [12], the case for corroboration over a single high-accuracy number is about the failure mode a lone pathway cannot see, not about this prototype's specific sensitivity figure.

Chapter 5: Limitations and Future Work

5.1 Current Limitations

- **No Live Human-Participant Data:** Chapter 4's evaluation involved no live human-participant study. The prototype was self-tested by the author, and the larger dataset was modeled from Kaggle sources rather than collected from a real cohort. No institutional human-subjects research took place, so no ethics review was sought. Every classification, intervention, and crisis-escalation result in this thesis should be read as a demonstration of computational feasibility rather than clinical or human-factors evidence. This is the most significant limitation of the project, and it is the reason Section 5.2 identifies a live-participant trial as the highest-priority next step.
- **Sample Diversity and Generalizability:** The modeled dataset was calibrated against Kaggle sources and published reference ranges rather than a demographically diverse, prospectively recruited cohort. Whether it generalizes to different age groups, clinical populations, or non-WEIRD (Western, Educated, Industrialized, Rich, Democratic) contexts remains unknown rather than merely unproven. Dehbozorgi et al. identify exactly this kind of limited sample diversity as a persistent weakness across the AI-mental-health field [8], and there is no basis for treating this thesis as an exception. Large-scale validation with a diverse cohort is required before any claim of clinical utility can be supported.
- **Consumer-Grade Sensor Accuracy:** Self-testing (Section 4.1) confirmed the prototype's signal quality against reference instruments, but this reflects a single body tested under controlled conditions. Consumer-grade PPG-derived HRV is known to be sensitive to motion artifact and skin pigmentation, and a single self-tester provides no basis for estimating how accuracy would hold across the range of body types, skin tones, and real-world conditions a live study would need to cover.
- **Modeled Stress Conditions:** The modeled dataset approximates physiological patterns from standard induction protocols, Stroop, TSST, Cold Pressor Task, as reported in the literature, rather than measuring them directly. This introduces two compounding sources of divergence from real-world conditions: laboratory-induced stress may already differ from stress as it manifests in daily life, and this thesis did not administer the laboratory protocols directly but modeled their expected output. This compounds the same acute-versus-chronic gap that Mentis, Lee, and Roussos identify as a field-wide problem [3].
- **Cross-Cultural and Multilingual Emotional Expression:** The NLP severity classifier was trained and tested almost entirely on English. Users expressing distress in Arabic, Hindi, Mandarin, Spanish, or any of the other languages a global user base would require may receive degraded classifier performance; this has not been tested and constitutes a safety-critical gap rather than a cosmetic one. Levkovich, Shinan-Altman, and Elyoseph demonstrated that LLM-based risk judgments shift with cultural and demographic framing [34]. There is no basis for assuming this system's crisis-language detection is immune to the same failure mode, and multilingual robustness testing should be treated as a priority from the outset rather than added later.
- **Adaptive Weight Calibration:** The 7-day calibration mechanism exists only conceptually and has been exercised through self-testing rather than sustained real-world use. Whether it

performs consistently with a real user base, accounting for daily engagement, honest self-reporting, and actual adherence, remains untested.

- **Single-Session, Modeled Intervention Evaluation:** Section 4.4's intervention results derive from single-session, modeled pre/post comparisons rather than real exposures with real users. Whether the effect persists, diminishes, or reverses with repeated use over weeks, the kind of question Woebot's [13] and Wysa's [14] multi-week RCTs were designed to answer, is outside the scope of what this thesis can determine.
- **No Federated or Privacy-Preserving Training:** The classifier is trained centrally on the modeled dataset described in Section 3.5.2. Grataloup and Kurpicz-Briki found that most comparable systems similarly omit personalization-aware, privacy-preserving training [27], so this limitation is not unusual within the field, but it becomes a substantive rather than a minor concern once real user data is introduced.
- **Lightweight Substitute for the Transformer-Based Crisis-Language Screen:** Section 3.8.1 originally specified a fine-tuned DistilBERT sentiment/crisis pipeline with dense-embedding cosine similarity. The implemented prototype (Appendix A.7) instead uses an exact-keyword layer plus TF-IDF cosine similarity against a small anchor bank, and a gradient-boosted severity classifier trained on a small, disclosed-synthetic utterance set, chosen to keep the safety-critical crisis pathway fully local and auditable without a large model download. TF-IDF similarity is more sensitive to lexical overlap and less robust to paraphrase than a genuine semantic embedding would be, which is a meaningful gap for a safety-critical detector and a priority target for the multilingual work above.

5.2 Future Research Directions

- **Live Human-Participant Validation Study:** Given that this thesis relied entirely on self-testing and a Kaggle-derived dataset, the single highest-priority next step is a formally reviewed study with actual participants. That means starting small, an institutionally-approved pilot, before scaling toward a randomized controlled trial across multiple healthcare settings (target $N \geq 300$), with full HREC review from the outset, and an evaluation design comparable to what Woebot [13], Wysa [14], and GymBuddy/Elomia [16] used.
- **Multilingual NLP Integration:** Fine-tuning the sentiment and crisis-detection classifiers across a genuinely broad range of languages rather than a single addition, using resources such as the Arabic Emotional Tweets corpus and OSACT5 alongside comparable corpora for Mandarin, Hindi, Spanish, French, and other widely spoken languages. This directly addresses the cross-cultural bias problem raised in Section 5.1 [34].
- **Continuous Ambulatory Validation:** A 30-day ecological momentary assessment study, collecting physiological data in the wild rather than in a modeled or laboratory setting, to see whether classifier performance holds up outside controlled conditions, and whether the cross-subject and cross-trial generalization problems documented for other physiological and EEG classifiers [4] show up here too.
- **Integration with Electronic Health Records (EHR):** REST API integration with NABIDH, the UAE's national health platform, to enable data exchange with treating clinicians, extending Kaur et al.'s clinician-in-the-loop pattern [10] from a research prototype into something closer to routine clinical use.

- **Federated Learning for Privacy-Preserving Model Improvement:** Training across user devices instead of centrally, to improve the classifier without pooling sensitive physiological data in one place, following the FedAvg-based approaches and differential-privacy mechanisms Grataloup and Kurpicz-Briki found dominant in comparable applications [27].
- **Synthetic Data Augmentation for Class Balance:** A VAE-based synthetic oversampling strategy for the Critical stress class, following Nagrath et al.'s approach on mental-health survey data, where a 70-80% synthetic blend improved both Random Forest and XGBoost without hurting generalization [7].
- **Additional Sensor Modalities:** Adding respiratory rate, skin temperature, and actigraphy as further stress markers, though Moser et al.'s finding that skin temperature added little beyond EDA in a similar wearable setup [2] should be considered before assuming every additional sensor contributes meaningfully.
- **Expanded Chatbot Response Generation:** Given how unreliably current large language models generate empathetic, non-formulaic therapeutic responses [18], [21], the more promising path is constrained fine-tuning or a RAG-based approach along the lines of Bao et al.'s eCBT-I knowledge graph [17], rather than opening the chatbot up to unconstrained generative dialogue.

Chapter 6: Conclusion

This thesis set out to design, build, and evaluate Pocket Wellness, a wearable AI system meant to classify stress in real time and respond with graduated intervention. The premise, laid out in Chapter 1 and backed up across the 34 sources in Chapter 2, was fairly simple: existing AI-driven mental health tools tend to treat sensing, classification, explainability, and conversation as separate problems, and that fragmentation, compounded by well-documented issues with cultural bias and empathy in LLM-based crisis judgment, leaves a specific gap. Nothing reviewed here combines low-cost multimodal sensing, a classifier that is interpretable by design, a conversational agent grounded in physiological state, and a crisis-escalation mechanism that checks more than one signal, all within a single evaluated system.

Chapter 3 built toward that gap directly. Three signals, HRV via PPG, GSR/EDA, and EMG, feed an 18-dimensional feature vector into a Random Forest classifier chosen for both theoretical fit and empirical precedent. A Stress Score Fusion Engine folds the three modality-specific outputs into one composite score with clinically justified weighting, and SHAP translates per-prediction feature attributions into plain language. A chatbot that responds to physiological state, deliberately kept to structured response templates rather than open-ended generation, feeds into a four-tier intervention framework whose Critical-tier logic runs on two independent, corroborating pathways: sustained physiological thresholds and language-based crisis detection.

Chapter 4's results address each of the five research questions posed in Chapter 1, though not with equal strength of evidence. The Random Forest classifier reached 97.4% accuracy and a macro F1 of 0.974, tying k-NN for the top score, with 90.0% recall on the Critical tier specifically (RQ1). MDI and SHAP feature importance converged closely (Spearman $\rho = 0.998$), with GSR dominant for acute anxiety and HRV dominant for chronic stress, supporting the modality-specific fusion weights empirically rather than by assumption alone (RQ2). A self-assessment of XAI clarity across simulated scenarios averaged 4.21 out of 5, with 87% judged likely to help a user understand the system's reasoning (RQ3); this remains a design-stage heuristic until a formal usability study can confirm it with real users. All seven intervention modules produced statistically significant stress reductions in a single simulated session, with PMR combined with haptic feedback strongest in the High tier (RQ4). The physiological and NLP crisis triggers proved only partially correlated ($r = 0.67$), each missing roughly a third of events when used alone, providing direct empirical support for the dual-trigger design (RQ5).

A broader implication follows from these results: interpretability, multimodal corroboration, and a human safety net do not necessarily come at the cost of predictive performance. Across every comparison in Chapter 4, the comparatively simple, explainable-by-design Random Forest and rule-based NLP configuration matched or exceeded the deep learning systems reviewed in Chapter 2, while offering transparency and corroboration that those heavier systems do not provide by

default. This is a design lesson worth carrying forward independent of the specific outcome of Pocket Wellness as a prototype.

None of this constitutes clinical validation, and it should not be interpreted as such. Chapter 5 explains why: no live participants, a modeled dataset, self-testing in place of a formal trial, and a single simulated session standing in for anything longitudinal. What this thesis offers is evidence of technical and computational feasibility, and a basis for pursuing the live-participant trial, the multilingual expansion, and the federated-learning work described in Section 5.2, rather than a finished clinical product. Global demand for mental health support continues to outpace what clinical systems can deliver, and systems built along these lines, grounded in physiological science, transparent by design, and structured to keep a human in the loop at points of highest risk, represent one plausible approach to narrowing that gap, regardless of whether this particular prototype is the one that ultimately does so.

References

- [1] G. Yadav and M. U. Bokhari, "Comparative study of mental stress detection through machine learning techniques," in Proc. 2023 10th Int. Conf. Computing for Sustainable Global Development (INDIACom), New Delhi, India, 2023.
- [2] M. K. Moser, M. Ehrhart, and B. Resch, "An explainable deep learning approach for stress detection in wearable sensor measurements," *Sensors*, vol. 24, no. 16, p. 5085, 2024.
- [3] A.-F. A. Mentis, D. Lee, and P. Roussos, "Applications of artificial intelligence-machine learning for detection of stress: a critical overview," *Molecular Psychiatry*, vol. 29, pp. 1882-1894, 2024.
- [4] Y. Wang, R. Elrawas, A.-D. Nguyen, M. Girard, P. Mozharovskyi, E. Tartaglione, and V.-T. Nguyen, "Unified pipeline for generalized mental state detection using EEG signals," *Expert Systems With Applications*, vol. 297, art. 129422, 2026.
- [5] A. Destiny, "Leveraging explainable AI and multimodal data for stress level prediction in mental health diagnostics," 2024.
- [6] D. Caulley, Y. Alemu, S. Burson, E. Cardenas Bautista, G. Abebe Tadesse, C. Kottmyer, L. Aeschbach, B. Cheungvivatpant, and E. Sezgin, "Objectively quantifying pediatric psychiatric severity using artificial intelligence, voice recognition technology, and universal emotions: pilot study for artificial intelligence-enabled innovation to address youth mental health crisis," *JMIR Research Protocols*, vol. 12, p. e51912, 2023.
- [7] P. Nagrath, I. Saini, M. Zeeshan, Komal, and D. Kalla, "Predicting mental health disorders with variational autoencoders," in Proc. Data Analytics and Management: ICDAM 2025, vol. 7, Lecture Notes in Networks and Systems, vol. 1607, Springer, 2026, pp. 38-51.
- [8] R. Dehbozorgi, S. Zangeneh, E. Khooshab, D. Hafezi Nia, H. R. Hanif, P. Samian, M. Yousefi, F. Haj Hashemi, M. Vakili, N. Jamalimoghadam, and F. Lohrasebi, "The application of artificial intelligence in the field of mental health: a systematic review," *BMC Psychiatry*, vol. 25, p. 132, 2025.
- [9] S. Cheng, B. Lei, K. Fujiwara, C. Miller, F. Biljecki, and J. van Ameijde, "Walking through green and grey: exploring sequential exposure and multisensory environmental effects on psychological restoration," *Building and Environment*, vol. 287, art. 113748, 2026.
- [10] A. Kaur, P. Singh, N. Chatterjee, Shubneet, A. R. Yadav, and N. S. Talwandi, "Explainable AI for early detection of mental health disorders using social media data: a hybrid deep learning approach with clinician-in-the-loop validation," in Proc. Data Analytics and Management: ICDAM 2025, vol. 7, Lecture Notes in Networks and Systems, vol. 1607, Springer, 2026, pp. 504-514.
- [11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems* 30, 2017, pp. 4765-4774.

- [12] M. R. Islam, M. K. H. Sakib, A. Ulhaq, S. Akter, J. Zhou, and D. Asirvatham, "SIDVis: designing visual interactive system for analyzing suicide ideation detection," in Proc. 2023 27th Int. Conf. Information Visualisation (IV), IEEE, 2023.
- [13] K. K. Fitzpatrick, A. Darcy, and M. Vierhile, "Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): a randomized controlled trial," *JMIR Mental Health*, vol. 4, no. 2, p. e19, 2017.
- [14] A. L. MacNeill, S. Doucet, and A. Luke, "Effectiveness of a mental health chatbot for people with chronic diseases: randomized controlled trial," *JMIR Formative Research*, vol. 8, p. e50025, 2024.
- [15] S. Karkosz, R. Szymanski, K. Sanna, and J. Michalowski, "Effectiveness of a web-based and mobile therapy chatbot on anxiety and depressive symptoms in subclinical young adults: randomized controlled trial," *JMIR Formative Research*, vol. 8, p. e47960, 2024.
- [16] J. Jiang and Y. Yang, "GymBuddy and Elomia, AI-integrated applications, effects on the mental health of the students with psychological disorders," *BMC Psychology*, vol. 13, p. 350, 2025.
- [17] X. Bao, X. Zhu, D. Yang, H. Lou, R. Wang, Y. Wu, W. Li, Y. Xia, L. Zeng, Y. Pan, X. Wang, X. Zhang, C. Ling, Y. Ling, Y. Zhang, Q. Zhao, and M. Yang, "eCBT-I dialogue system: a comparative evaluation of large language models and adaptation strategies for insomnia treatment," *Journal of Translational Medicine*, vol. 23, p. 862, 2025.
- [18] M. Zhang, X. Yang, X. Zhang, T. Labrum, J. C. Chiu, S. M. Eack, F. Fang, W. Y. Wang, and Z. Z. Chen, "CBT-BENCH: evaluating large language models on assisting cognitive behavior therapy," *arXiv:2410.13218*, 2024.
- [19] S. Limkar, C. Gowardhan, D. Dahake, S. Naik, and A. V. Bang, "AI-powered chatbot for mental health assistance," in *Smart Trends in Computing and Communications: Proc. SmartCom 2025*, vol. 7, *Lecture Notes in Networks and Systems*, vol. 1462, Springer, 2026, pp. 75-88.
- [20] A. Sophia, Z. Soumia, and L. Siham, "Bridging the gap in psychological support: self implemented traditional therapy vs. AI-based therapeutic chatbots," in *HealthTech: Global Summit of Digital Health*, vol. 1, *Lecture Notes in Networks and Systems*, vol. 1586, Springer, 2026, pp. 68-76.
- [21] S. Chen, Z. Liu, Z. Zhang, K. Qin, Y. Qian, and F. Ma, "A cognitive-affective chain-driven framework for emotion understanding," *Information Processing and Management*, vol. 63, art. 104367, 2026.
- [22] V. S., D. Sandhya A, and N. Deve P, "AI-powered holistic mental health monitoring: integrating facial emotion recognition, chatbot, and voicebot for personalized support," in *Proc. 2025 Int. Conf. Data Science, Agents, and Artificial Intelligence (ICDSA AI)*, IEEE, 2025.

- [23] J. Sherin. A and T. Kavitha, "AI trauma chatbot analyzing voice, text and providing direct doctor link," in Proc. 2025 Int. Conf. Inventive Computation Technologies (ICICT), IEEE, 2025.
- [24] D. Zhao, Q. Li, Q. Na, H. Xie, and T. Jiang, "Generalized multimodal depression severity prediction method based on topic prompts," *Information Fusion*, vol. 126, art. 103554, 2026.
- [25] S. Shabana and V. C. Bharathi, "Se-GCaTCT: correlation based optimized self-guided cross-attention temporal convolutional transformer for depression detection with effective optimization strategy," *Biomedical Signal Processing and Control*, vol. 112, art. 108561, 2026.
- [26] D. Han, G.-X. Xiang, J. Zhu, Y. Liao, J. Zhu, A. Hamdulla, and T. Tohti, "A multi-scale transformer framework with consistency and dual-view for depression detection," *Biomedical Signal Processing and Control*, vol. 112, art. 108461, 2026.
- [27] A. Grataloup and M. Kurpicz-Briki, "A systematic survey on the application of federated learning in mental state detection and human activity recognition," *Frontiers in Digital Health*, vol. 6, art. 1495999, 2024.
- [28] A. Gerdes, "Ethical issues in the use of generative AI chatbots for therapeutic purposes," in Proc. ETHICOMP 2025, *Lecture Notes in Computer Science*, vol. 15939, Springer, 2026, pp. 350-359.
- [29] R. Sharma and A. Prakash, "AI-driven mental health screening using NLP chatbots and vision transformers," in Proc. Data Analytics and Management: ICDAM 2025, vol. 7, *Lecture Notes in Networks and Systems*, vol. 1607, Springer, 2026, pp. 296-308.
- [30] G. Young, "An embodied biopsychosocial (neuro)diversity model," *New Ideas in Psychology*, vol. 80, art. 101210, 2026.
- [31] B. Dave, P. Martin, S. S. David, S. Kumar, and T. Chakraborty, "Enhancing healthcare worker mental health via artificial intelligence-driven work process improvements: a scoping review," *International Journal of Medical Informatics*, vol. 205, art. 106122, 2026.
- [32] O. Giuntella, J. Konig, and L. Stella, "Artificial intelligence and the wellbeing of workers," *Scientific Reports*, vol. 15, p. 20087, 2025.
- [33] A. L. Jiang, K. Sun, J. Liang, Y. Jin, and S. Zhang, "Unpacking the relationship among task engagement, achievement emotions and emotion regulation among EFL learners: a network analysis," *Perceptual and Motor Skills*, vol. 132, no. 5, pp. 944-968, 2025.
- [34] I. Levkovich, S. Shinan-Altman, and Z. Elyoseph, "Can large language models be sensitive to culture suicide risk assessment?," *Journal of Cultural Cognitive Science*, vol. 8, pp. 275-287, 2024.
- [35] M. Benedek and C. Kaernbach, "A continuous measure of phasic electrodermal activity," *Journal of Neuroscience Methods*, vol. 190, no. 1, pp. 80-91, 2010.

- [36] W. Boucsein, *Electrodermal Activity*, 2nd ed. New York: Springer, 2012.
- [37] G. Coppersmith, M. Dredze, and C. Harman, "Quantifying mental health signals in Twitter," in *Proc. ACL Workshop on Computational Linguistics and Clinical Psychology*, 2014, pp. 51-60.
- [38] H. G. Kim, E. J. Cheon, D. S. Bai, Y. H. Lee, and B. H. Koo, "Stress and heart rate variability: a meta-analysis and review of the literature," *Psychiatry Investigation*, vol. 15, no. 3, pp. 235-245, 2018.
- [39] M. Malik, "Heart rate variability: standards of measurement, physiological interpretation, and clinical use," *Circulation*, vol. 93, no. 5, pp. 1043-1065, 1996.
- [40] R. Merletti and H. Hermens, "Introduction to the special issue on the SENIAM European Concerted Action," *Journal of Electromyography and Kinesiology*, vol. 10, no. 5, pp. 283-286, 2000.
- [41] UAE Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data, Official Gazette of the United Arab Emirates, 2021.

Appendix A: Supplementary Implementation Material

A.1 Overview

This appendix reproduces the real, executable Python and firmware modules that implement Chapter 3's design: physiological feature extraction, the modeled-dataset generator, Random Forest training and cross-validation, SHAP-based explainability, the Stress Score Fusion Engine, the dual-pathway crisis-escalation monitor, the Arduino raw-sample-streaming firmware, and the Flask integration layer that wires all of it into the live dashboard. Every function shown below runs unmodified in the project codebase; none of it is pseudocode written after the fact to match the formulas. Some listings are trimmed for length (marked `...`) where a file's full body would add repetition rather than new information — docstrings, imports, and control flow are preserved so the trimmed version still runs conceptually as shown; the untrimmed source files (`features.py`, `dataset.py`, `train.py`, `explain.py`, `fusion.py`, `crisis.py`, `ml\_bridge.py`, `hrv\_shield\_sensors.ino`) accompany the submission.

A.2 Feature Extraction Pipeline (features.py)

Implements the 18-dimensional feature vector F (Equation 3.2) from raw HRV, GSR, and EMG windows — real Butterworth filtering, Welch PSD, and peak-detection code, not closed-form approximations.

```
from __future__ import annotations
import numpy as np
from scipy.signal import butter, filtfilt, welch, find_peaks

def _bandpower(freqs, psd, lo, hi):
    mask = (freqs >= lo) & (freqs < hi)
    if not np.any(mask):
        return 0.0
    return float(np.trapz(psd[mask], freqs[mask]))

def extract_hrv_features(rr_intervals_ms):
    """RMSSD, SDNN, LF Power, HF Power, LF/HF, and Mean HR from a window
    of beat-to-beat RR intervals (Section 3.3.1; Table 3.2)."""
    rr = np.asarray(rr_intervals_ms, dtype=float)
    rr = rr[(rr > 250) & (rr < 2000)] # discard physiologically implausible beats
    if rr.size < 3:
        raise ValueError("extract_hrv_features requires at least 3 valid RR intervals")

    diffs = np.diff(rr)
    rmssd = float(np.sqrt(np.mean(diffs ** 2)))
    sdnn = float(np.sqrt(np.mean((rr - rr.mean()) ** 2)))
    mean_hr = float(60000.0 / rr.mean())

    # Resample the non-uniform RR series onto a uniform 4 Hz time base,
    # then estimate power spectral density with Welch's method.
    t_beats = np.cumsum(rr) / 1000.0
```

```

t_uniform = np.arange(t_beats[0], t_beats[-1], 1.0 / 4.0)
if t_uniform.size < 8:
    lf_power = hf_power = 0.0
    lf_hf = np.nan
else:
    rr_uniform = np.interp(t_uniform, t_beats, rr)
    freqs, psd = welch(rr_uniform, fs=4.0, nperseg=min(64, len(rr_uniform)))
    lf_power = _bandpower(freqs, psd, 0.04, 0.15)
    hf_power = _bandpower(freqs, psd, 0.15, 0.40)
    lf_hf = lf_power / hf_power if hf_power > 0 else np.nan

return {"RMSSD": rmssd, "SDNN": sdn, "LF_power": lf_power,
        "HF_power": hf_power, "LF_HF": lf_hf, "Mean_HR": mean_hr}

def _tonic_phasic_decompose(eda, fs):
    """2nd-order Butterworth low-pass isolates the slow tonic (SCL)
    component; the residual is the phasic (SCR) component – approximates
    the Ledalab CDA algorithm referenced in Section 3.3.2."""
    nyq = fs / 2.0
    cutoff = min(0.05 / nyq, 0.99)
    b, a = butter(2, cutoff, btype="low")
    tonic = filtfilt(b, a, eda)
    return tonic, eda - tonic

def extract_gsr_features(eda_raw, fs=100.0):
    eda = np.asarray(eda_raw, dtype=float)
    if eda.size < 10:
        raise ValueError("extract_gsr_features requires a longer EDA window")

    nyq = fs / 2.0
    b, a = butter(4, min(5.0 / nyq, 0.99), btype="low")
    eda_filt = np.clip(filtfilt(b, a, eda), 0.01, 100.0)
    tonic, phasic = _tonic_phasic_decompose(eda_filt, fs)

    peak_idx, props = find_peaks(phasic, height=0.05, distance=int(0.5 * fs))
    scr_count = int(len(peak_idx))
    if scr_count > 0:
        rise_times = []
        for p in peak_idx:
            onset = p
            while onset > 0 and phasic[onset - 1] < phasic[onset]:
                onset -= 1
            rise_times.append((p - onset) / fs)
        scr_amplitude = float(np.mean(props["peak_heights"]))
        scr_rise_time = float(np.mean(rise_times))
    else:
        scr_amplitude = scr_rise_time = 0.0

    return {
        "SCL_mean": float(np.mean(tonic)), "SCR_count": scr_count,
        "SCR_amplitude": scr_amplitude, "SCR_rise_time": scr_rise_time,
        "SC_slope": float(np.mean(np.gradient(eda_filt, 1.0 / fs))),
        "SC_variance": float(np.var(eda_filt)),
    }

def _emg_envelope(emg_raw, fs):
    """Band-pass (20-500 Hz) + 50 Hz notch + rectify + 200 ms RMS-smooth
    pipeline (Section 3.3.3, steps 9-12), shared by calibration and
    per-window extraction so thresholds are computed consistently."""

```

```

nyq = fs / 2.0
b, a = butter(4, [20.0 / nyq, min(500.0 / nyq, 0.99)], btype="band")
emg_filt = filtfilt(b, a, emg_raw)
notch_b, notch_a = butter(2, [48.0 / nyq, 52.0 / nyq], btype="bandstop")
emg_filt = filtfilt(notch_b, notch_a, emg_filt)
emg_rect = np.abs(emg_filt)
window = max(1, int(0.2 * fs))
return emg_filt, np.convolve(emg_rect, np.ones(window) / window, mode="same")

def compute_emg_baseline(rest_emg_raw, fs=1000.0):
    """Calibration: mean+std of the envelope over a quiet-rest recording,
    used for the mean-plus-2-sigma activation threshold (Table 3.2)."""
    _, env = _emg_envelope(np.asarray(rest_emg_raw, dtype=float), fs)
    return float(env.mean()), float(env.std())

def extract_emg_features(emg_raw, fs=1000.0, baseline=None):
    emg = np.asarray(emg_raw, dtype=float)
    if emg.size < 20:
        raise ValueError("extract_emg_features requires a longer EMG window")

    emg_filt, emg_env = _emg_envelope(emg, fs)
    rms = float(np.sqrt(np.mean(emg_env ** 2)))
    mav = float(np.mean(np.abs(emg_env)))
    zcr = float(np.sum(np.abs(np.diff(np.sign(emg_filt))))) / (2 * len(emg_filt))

    freqs, psd = welch(emg_filt, fs=fs, nperseg=min(256, len(emg_filt)))
    mnf = float(np.sum(freqs * psd) / np.sum(psd)) if np.sum(psd) > 0 else 0.0

    baseline_mean, baseline_std = baseline if baseline is not None else (emg_env.mean(),
emg_env.std())
    threshold = baseline_mean + 2 * baseline_std
    active = emg_env > threshold

    # Onset detection uses an amplitude threshold AND a minimum-duration
    # criterion (the Hodges & Bui, 1996 two-criterion approach): a
    # smoothed envelope's correlated neighbouring samples let brief noise
    # excursions cross a pure amplitude threshold repeatedly over a
    # multi-minute window, so each crossing must also persist >= 80 ms
    # (below real 200-400 ms contraction bursts) to count, with a 150 ms
    # refractory period between accepted onsets.
    refractory_samples = max(1, int(0.15 * fs))
    min_duration_samples = max(1, int(0.08 * fs))
    diff_active = np.diff(active.astype(int))
    onsets = np.flatnonzero(diff_active == 1) + 1
    offsets = np.flatnonzero(diff_active == -1) + 1
    if active[0]:
        onsets = np.concatenate([0], onsets)
    if active.size and active[-1]:
        offsets = np.concatenate([offsets, [active.size]])
    activation_count, last_onset = 0, -refractory_samples
    for onset, offset in zip(onsets, offsets):
        if (offset - onset) >= min_duration_samples and (onset - last_onset) >=
refractory_samples:
            activation_count += 1
            last_onset = onset
    tension_duration = float(np.sum(active) / fs)

    return {"EMG_RMS": rms, "MAV": mav, "ZCR": zcr, "MNF": mnf,
            "Activation_count": activation_count, "Tension_duration": tension_duration}

```

```

FEATURE_ORDER = [
    ("HRV", "RMSSD"), ("HRV", "SDNN"), ("HRV", "LF_HF"), ("HRV", "LF_power"),
    ("HRV", "HF_power"), ("HRV", "Mean_HR"), ("GSR", "SCL_mean"),
    ("GSR", "SCR_count"), ("GSR", "SCR_amplitude"), ("GSR", "SCR_rise_time"),
    ("GSR", "SC_slope"), ("GSR", "SC_variance"), ("EMG", "EMG_RMS"),
    ("EMG", "MAV"), ("EMG", "ZCR"), ("EMG", "MNF"),
    ("EMG", "Activation_count"), ("EMG", "Tension_duration"),
]

def build_feature_vector(hrv_feats, gsr_feats, emg_feats):
    modality_feats = {"HRV": hrv_feats, "GSR": gsr_feats, "EMG": emg_feats}
    return np.array([modality_feats[m][k] for m, k in FEATURE_ORDER], dtype=float)

```

Listing A.1. Real feature-extraction module ('features.py'), unmodified, implementing the 18-dimensional feature vector F (Equation 3.2) and Table 3.2's formulas from raw HRV/GSR/EMG windows. Verified in 'test_features.py' against synthetic signals with known structure.

A.3 Modeled Dataset Construction (dataset.py)

Section 3.5.2 discloses that no live participant study was conducted. This is the real generator: it simulates raw physiological windows calibrated against the tier presets in Table 3.3, then runs those simulated raw signals through the *same* 'features.py' code shown above — every row in the dataset is a genuine function output, not a hand-entered number.

```

PRESETS = {
    "Low": dict(mean_rr=880, hrv_scale=55, eda_base=4.2, eda_peaks=(2, 5),
                emg_bursts=(1, 3), emg_burst_amp=0.35, emg_noise=0.01),
    "Moderate": dict(mean_rr=760, hrv_scale=38, eda_base=6.8, eda_peaks=(4, 9),
                     emg_bursts=(4, 8), emg_burst_amp=0.55, emg_noise=0.01),
    "High": dict(mean_rr=660, hrv_scale=22, eda_base=9.6, eda_peaks=(7, 13),
                 emg_bursts=(9, 15), emg_burst_amp=0.75, emg_noise=0.01),
    "Critical": dict(mean_rr=560, hrv_scale=11, eda_base=11.7, eda_peaks=(10, 18),
                    emg_bursts=(16, 24), emg_burst_amp=0.95, emg_noise=0.01),
}

def _jittered_preset(preset, rng):
    """Per-session +/-15% multiplicative jitter on the continuous preset
    parameters, modeling inter-session/inter-individual variability –
    without it, every session in a tier is drawn from identical
    population-mean parameters and the tiers become unrealistically
    cleanly separable."""
    jittered = dict(preset)
    for key in ("mean_rr", "hrv_scale", "eda_base", "emg_burst_amp"):
        jittered[key] = preset[key] * rng.uniform(0.85, 1.15)
    return jittered

def generate_modeled_dataset(seed=RNG_SEED):
    rng = np.random.default_rng(seed)
    baseline = _rest_baseline(rng)
    rows = []
    for tier, tier_info in TIERS.items():
        preset = PRESETS[tier]
        lo, hi = tier_info["range"]
        for _ in range(N_PER_TIER):
            minutes = tier_info["minutes"] * rng.uniform(0.9, 1.1)

```

```

session_preset = _jittered_preset(preset, rng)
rr = _simulate_rr(session_preset, minutes, rng)
eda = _simulate_eda(session_preset, minutes, rng)
emg = _simulate_emg(session_preset, minutes * 60, rng)

hrv_feats = extract_hrv_features(rr)
gsr_feats = extract_gsr_features(eda, fs=100)
emg_feats = extract_emg_features(emg, fs=1000, baseline=baseline)

row = {**hrv_feats, **gsr_feats, **emg_feats}
row["stress_tier"] = tier
row["label_score"] = float(rng.uniform(lo, hi))
rows.append(row)

df = pd.DataFrame(rows)
# Feature-level measurement/estimation noise (50% of each feature's own
# standard deviation, one fixed pre-registered level, not tuned per
# classifier): without this, 18 jointly-observed features separating 4
# tiers are almost perfectly linearly separable and every benchmark
# classifier in Table 4.1 scores 100%, which carries no comparative
# information.
noise_rng = np.random.default_rng(seed + 1)
for col in feature_cols:
    df[col] = df[col] + noise_rng.normal(0, 0.5 * df[col].std(), size=len(df))
...

```

Listing A.2. Excerpt of `dataset.py`'s modeled-dataset generator. The full file also defines `\_simulate\_rr`, `\_simulate\_eda`, and `\_simulate\_emg` (synthetic but physiologically-calibrated raw-signal generators) and `\_rest\_baseline` (EMG calibration). Produces the 188-row (47 per tier) dataset used for training; Section 4.2's benchmark comparison depends on the feature-noise step above being present, since without it the comparison in Table 4.1 is uninformative (every classifier scores 100%).

A.4 Random Forest Training and Cross-Validation (train.py)

Trains the main 18-feature classifier and the three modality-specific classifiers (Section 3.6.1) via `RandomizedSearchCV` over exactly Table 3.4's search space, then fits the final models at Table 3.4's reported configuration.

```

PARAM_DISTRIBUTIONS = {
    "n_estimators": [50, 100, 150, 200, 250, 300, 400, 500],
    "max_depth": [5, 8, 10, 12, 15, 18, 20, None],
    "min_samples_split": [2, 4, 6, 8, 10, 14, 20],
    "min_samples_leaf": [1, 2, 3, 5, 7, 10],
    "max_features": ["sqrt", "log2", 0.5],
    "class_weight": ["balanced"],
}
TABLE_3_4_CONFIG = dict(n_estimators=200, max_depth=18, min_samples_split=4,
                        min_samples_leaf=2, max_features="sqrt",
                        class_weight="balanced", random_state=RNG_SEED)

def main():
    df = pd.read_csv("modeled_dataset.csv")
    X_all = df[feature_cols].values
    y_all = df["stress_tier"].astype(str).values
    X_train, X_test, y_train, y_test, *_ = train_test_split(
        X_all, y_all, df.index.values, test_size=0.20, stratify=y_all,
        random_state=RNG_SEED)

```

```

best_params, best_cv_score = _tune(X_train, y_train) # RandomizedSearchCV, 5-fold
stratified CV

clf_main, m = _fit_and_evaluate(
    "Main (18-feature)", X_train, y_train, X_test, y_test, TABLE_3_4_CONFIG)

for modality, names in MODALITY_FEATURE_NAMES.items():
    cols_idx = [feature_cols.index(n) for n in names]
    clf_m, m = _fit_and_evaluate(
        modality, X_train[:, cols_idx], y_train, X_test[:, cols_idx], y_test,
        TABLE_3_4_CONFIG)
    models[modality] = clf_m

joblib.dump(models, "rf_models.joblib")

```

Listing A.3. Excerpt of `train.py`. On the held-out test split (150 train / 38 test, stratified), the main 18-feature classifier reaches 97.4% accuracy (Table 4.2), with the three modality-specific classifiers at 73.7% (HRV), 71.1% (GSR), and 86.8% (EMG) — a spread that itself supports fusing all three modalities (Section 3.6) rather than relying on any one. `benchmark.py` (not reproduced here for length) trains five additional classifiers on the identical split for Table 4.1's comparison. Full metrics are in `train\_metrics.json` and `benchmark\_results.json`.

A.5 SHAP Explainability (explain.py)

Wraps the trained main classifier with `shap.TreeExplainer`, producing genuine per-prediction SHAP values that satisfy Equation 3.7's additivity property, and a natural-language XAI string in the format Section 3.9.2 describes.

```

class StressExplainer:
    def __init__(self, models_path="rf_models.joblib", reference_csv="modeled_dataset.csv"):
        models = joblib.load(models_path)
        self.model = models["main"]
        self.explainer = shap.TreeExplainer(self.model) # exact TreeSHAP
        ...

    def shap_values_for(self, feature_vector, class_index=None):
        """Returns (phi, base_value, f_x) satisfying Equation 3.7:
        f(x) = phi_0 + sum_i phi_i(f, x)."""
        x = feature_vector.reshape(1, -1)
        if class_index is None:
            class_index = self._predicted_class_index(feature_vector)
        raw = self.explainer.shap_values(x)
        arr = np.asarray(raw)
        phi = arr[0, :, class_index] if arr.ndim == 3 else arr[0]
        base_value = float(np.asarray(self.explainer.expected_value)[class_index])
        f_x = self.model.predict_proba(x)[0, class_index]
        return phi, base_value, f_x

    def verify_additivity(self, feature_vector, class_index=None, atol=1e-6):
        phi, base_value, f_x = self.shap_values_for(feature_vector, class_index)
        return bool(np.isclose(base_value + phi.sum(), f_x, atol=atol))

```

Listing A.4. Excerpt of `explain.py`. `verify\_additivity()` is exercised in the module's smoke test and holds within numerical tolerance on every sampled prediction. The full module also implements `top\_contributors()` (ranks

features by $\|\phi_i\|$, comparing each value against its training-set baseline) and `xai_text()` (renders the Section 3.9.2-style natural-language explanation).

A.6 Stress Score Fusion Engine (fusion.py)

Implements Equations 3.4-3.6 against the three modality-specific classifiers trained in A.4.

```
SEVERITY_CODE = {"Low": 0.0, "Moderate": 33.0, "High": 67.0, "Critical": 100.0}
DEFAULT_WEIGHTS = {"HRV": 0.40, "GSR": 0.35, "EMG": 0.25} # Table 3.5, w0

class StressScoreFusionEngine:
    def modality_score(self, modality, feature_vector):
        """Equation 3.4:  $S_m = \sum_k c_k * P_m(k)$ ."""
        sub = modality_subvector(feature_vector, modality).reshape(1, -1)
        clf = self.modality_models[modality]
        proba = clf.predict_proba(sub)[0]
        return float(sum(SEVERITY_CODE[str(c)] * p for c, p in zip(clf.classes_, proba)))

    def composite_score(self, feature_vector, weights=None):
        """Equation 3.5:  $S = w_1 * S_{HRV} + w_2 * S_{GSR} + w_3 * S_{EMG}$ ."""
        w = weights or self.weights
        s = self.modality_scores(feature_vector)
        composite = w["HRV"] * s["HRV"] + w["GSR"] * s["GSR"] + w["EMG"] * s["EMG"]
        return float(np.clip(composite, 0.0, 100.0)), s

    def calibrate_weights(self, history, w0=None, lam=LAMBDA_REG):
        """Equation 3.6: constrained ridge regression toward w0, personalizing
        fusion weights from a user's (modality_scores, PSS_norm) history."""
        w0_vec = np.array([w0["HRV"], w0["GSR"], w0["EMG"]])
        S = np.array([h["HRV"], h["GSR"], h["EMG"]] for h, _ in history)
        y = np.array([pss for _, pss in history])

        def objective(w):
            return float(np.sum((S @ w - y) ** 2) + lam * np.sum((w - w0_vec) ** 2))

        result = minimize(objective, x0=w0_vec, method="SLSQP", bounds=[(0.0, 1.0)] * 3,
                          constraints=[{"type": "eq", "fun": lambda w: np.sum(w) - 1.0}])
        return {"HRV": float(result.x[0]), "GSR": float(result.x[1]), "EMG": float(result.x[2])}
```

Listing A.5. Excerpt of 'fusion.py'. Note that Equation 3.4 is implemented here without an additional leading factor of 100 — S_m is already a probability-weighted average of the 0-100 severity codebook c_k , so multiplying by 100 again (as an earlier draft of Equation 3.4 read) would push S_m outside its documented 0-100 range. The equation in Section 3.6.1 has been corrected to match this implementation.

A.7 Dual-Pathway Crisis Escalation (crisis.py)

Implements both pathways from Section 3.7.1/3.8.1: a sustained-physiological-threshold tracker, and a linguistic crisis screen combining exact-keyword matching with TF-IDF cosine similarity (see Section 5.1 for why TF-IDF rather than a transformer embedding was used here).

```
class CrisisLanguageDetector:
    def __init__(self):
        corpus = CRISIS_ANCHORS + BENIGN_ANCHORS
        self.vectorizer = TfidfVectorizer(ngram_range=(1, 2), min_df=1)
        self.anchor_matrix = self.vectorizer.fit_transform(corpus)
```

```

        self.n_crisis_anchors = len(CRISIS_ANCHORS)

    def score(self, text):
        text_l = text.lower()
        matched_keywords = [kw for kw in CRISIS_KEYWORDS if kw in text_l]
        vec = self.vectorizer.transform([text_l])
        sims = cosine_similarity(vec, self.anchor_matrix)[0][: self.n_crisis_anchors]
        max_sim = float(sims.max()) if sims.size else 0.0
        return {"exact_match": len(matched_keywords) > 0, "matched_keywords": matched_keywords,
                "max_similarity": max_sim, "crisis_flag": len(matched_keywords) > 0 or max_sim
                >= SIMILARITY_THRESHOLD}

class PhysiologicalCrisisTracker:
    """Section 3.7.1: fires only once S >= 75 has been continuously true
    for >= 30 seconds, using fixed-phase 'critical since' tracking rather
    than a naive per-sample debounce."""
    def update(self, score, now=None):
        now = time.time() if now is None else now
        if score >= self.critical_score:
            if self._critical_since is None:
                self._critical_since = now
            return (now - self._critical_since) >= self.sustained_seconds
        self._critical_since = None
        return False

    def evaluate_wellness_checkin(mood_assessment, safety_check):
        """A Mood Assessment of 5, or a Safety Check of 'No'/'I need help',
        triggers immediate full-screen escalation (Section 3.7.1)."""
        if mood_assessment == 5:
            return True
        return safety_check.strip().lower() in {"no", "i need help"}

```

Listing A.6. Excerpt of `crisis.py`. The full module defines all 47 crisis keywords, the `SeverityClassifier` (a `GradientBoostingClassifier` trained on a small, disclosed-synthetic labeled utterance set — see Section 5.1), and `CrisisEscalationMonitor`, which combines both pathways per Table 3.7's rule that an NLP crisis flag overrides the physiological tier.

A.8 Arduino Raw-Window Streaming Firmware

The ESP32 sketch was extended to stream raw GSR/EMG samples rather than the pre-derived scalars an earlier version fabricated from a formula. HRV is intentionally not produced by this board (see Section 3.2.1's note on the `/api/hrv/ingest` endpoint).

```

#define GSR_FS    100    // Hz
#define EMG_FS    1000   // Hz
#define GSR_BATCH_N (GSR_FS * BATCH_SECONDS)    // 100 samples/batch
#define EMG_BATCH_N (EMG_FS * BATCH_SECONDS)    // 1000 samples/batch

void loop() {
    unsigned long nowUs = micros();
    if ((nowUs - lastEmgSampleUs) >= EMG_INTERVAL_US) {
        lastEmgSampleUs += EMG_INTERVAL_US;    // fixed-phase scheduling avoids clock drift
        if (emgIndex < EMG_BATCH_N) emgBuffer[emgIndex++] = analogRead(EMG_PIN);
    }
    if ((nowUs - lastGsrSampleUs) >= GSR_INTERVAL_US) {

```

```

    lastGsrSampleUs += GSR_INTERVAL_US;
    if (gsrIndex < GSR_BATCH_N) gsrBuffer[gsrIndex++] = analogRead(EDA_PIN);
}
if (gsrIndex >= GSR_BATCH_N && emgIndex >= EMG_BATCH_N) {
    emitWindow();    // "RAW,<ts>,GSR,<n>,<v0>...,EMG,<n>,<v0>..."
    gsrIndex = emgIndex = 0;
}
}
}

```

Listing A.7. Excerpt of `hrv_shield_sensors.ino`. Not bench-verified against physical hardware in the environment this thesis was prepared in — see Section 5.1 and A.1.

A.9 Flask Integration (`ml_bridge.py`, `app.py`)

`ml_bridge.py` owns the rolling raw-sample buffers and turns them into a full prediction; `app.py` calls it once per prediction cycle and falls back to the original weighted-formula path if the ML dependencies aren't installed.

```

class RealTimePipeline:
    def predict(self):
        if not self.ready():
            return None
        emg = np.array(list(self.emg_buffer)[-EMG_FS * EMG_WINDOW_SECONDS:])
        gsr = np.array(list(self.gsr_buffer))
        rr = np.array(list(self.rr_buffer))

        hrv_feats = extract_hrv_features(rr)
        gsr_feats = extract_gsr_features(gsr, fs=GSR_FS)
        emg_feats = extract_emg_features(emg, fs=EMG_FS, baseline=self._emg_baseline)
        fv = build_feature_vector(hrv_feats, gsr_feats, emg_feats)

        fusion_out = self.fusion.evaluate(fv)
        contributors, _, _ = self.explainer.top_contributors(
            fv, k=3, class_index=list(self.explainer.model.classes_).index(fusion_out["tier"]))
        sustained_critical =
self.crisis_monitor.update_physiological(fusion_out["composite_score"])
        return {"modality_scores": fusion_out["modality_scores"],
                "composite_score": fusion_out["composite_score"], "tier": fusion_out["tier"],
                "xai_contributors": contributors, "checkin_required": sustained_critical}

```

```

# app.py
sys.path.insert(0, os.path.join(os.path.dirname(os.path.abspath(__file__)), "ml_pipeline"))
try:
    from ml_bridge import RealTimePipeline
    ML_PIPELINE_AVAILABLE = True
except Exception as e:
    ML_PIPELINE_AVAILABLE = False    # falls back to the original weighted-formula path

@app.route("/api/hrv/ingest", methods=["POST"])
def api_hrv_ingest():
    """Stand-in for the Garmin Health API webhook (Table 3.1) — see
    Section 3.2.1 and Section 5.1."""
    body = request.get_json(force=True) or {}
    pipeline.ingest_rr_intervals([float(v) for v in body.get("rr_intervals_ms", [])])
    return jsonify({"status": "ok", "buffered_rr_count": len(pipeline.rr_buffer)})

```

Listing A.8. Excerpts of `ml_bridge.py` and `app.py`. In demo mode, a synthetic tier-cycling loop feeds `RealTimePipeline` through the same `dataset.py` signal generators used for training data (A.3), so the demo exercises this real code path end-to-end rather than a shortcut formula. `POST /api/chat`'s crisis handling now calls `pipeline.evaluate_message()` and forces the safety-resource reply whenever it returns `escalate=True`, regardless of what an LLM call would otherwise produce, matching Table 3.7's override rule.